



Slitting Process Technology

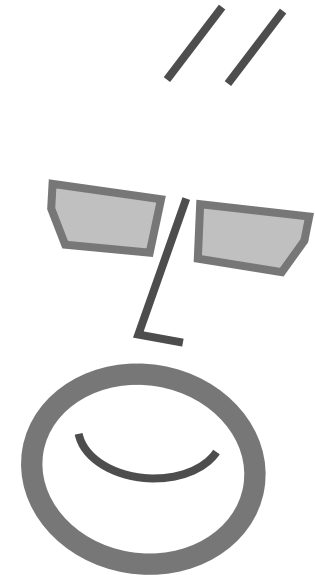
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Slitting Qs

How would you cut _____?



How Would You Cut These?



How Would You Cut These?



Brittle
Thin



Firm
Dry



Soft
Sticky



Hard/Soft
/Hard
Thin



Brittle
Sticky
Thin

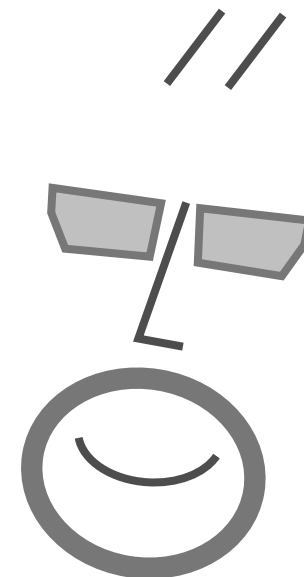


Strong
Abrasive
Thin



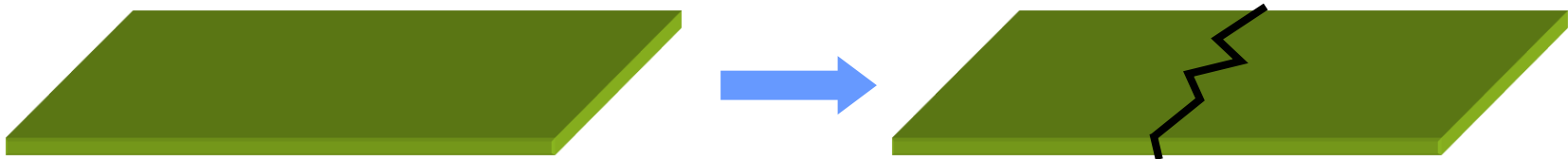
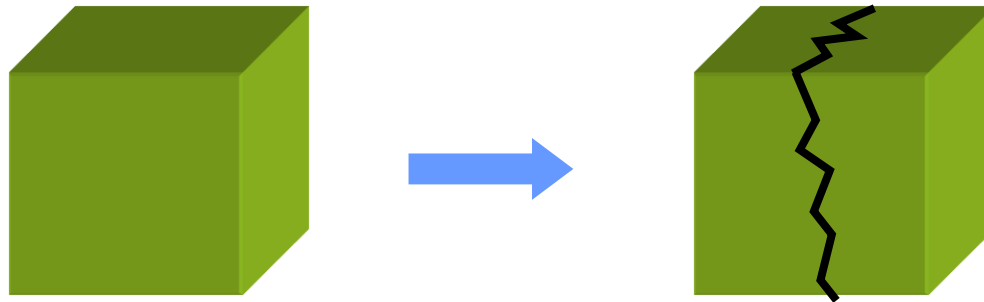
Slitting Qs

How is thinking about stress important to slitting?



Fracture Mechanics

Why do things break?

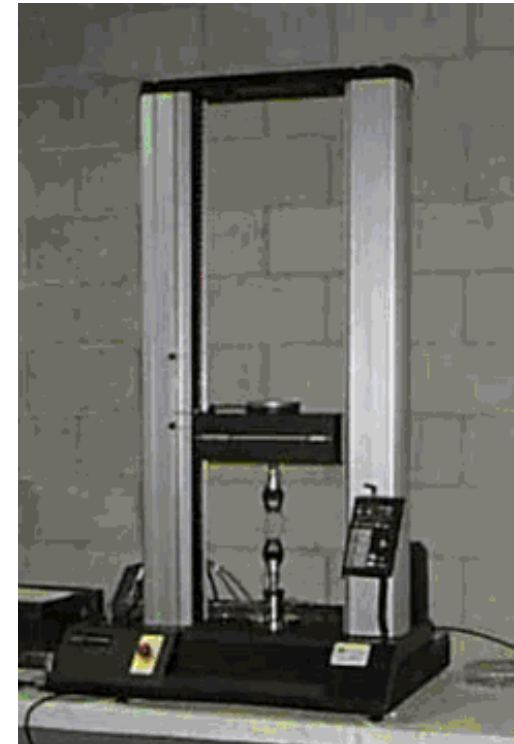
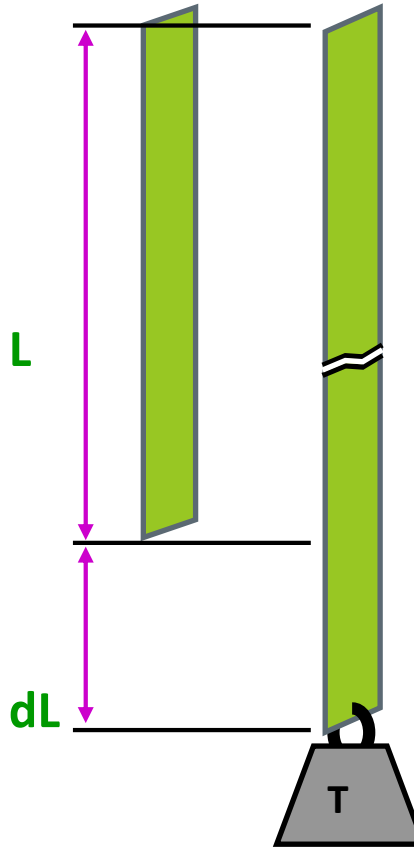


Break Testing

In tensile-elongation testing, a sample is elongated until it breaks.

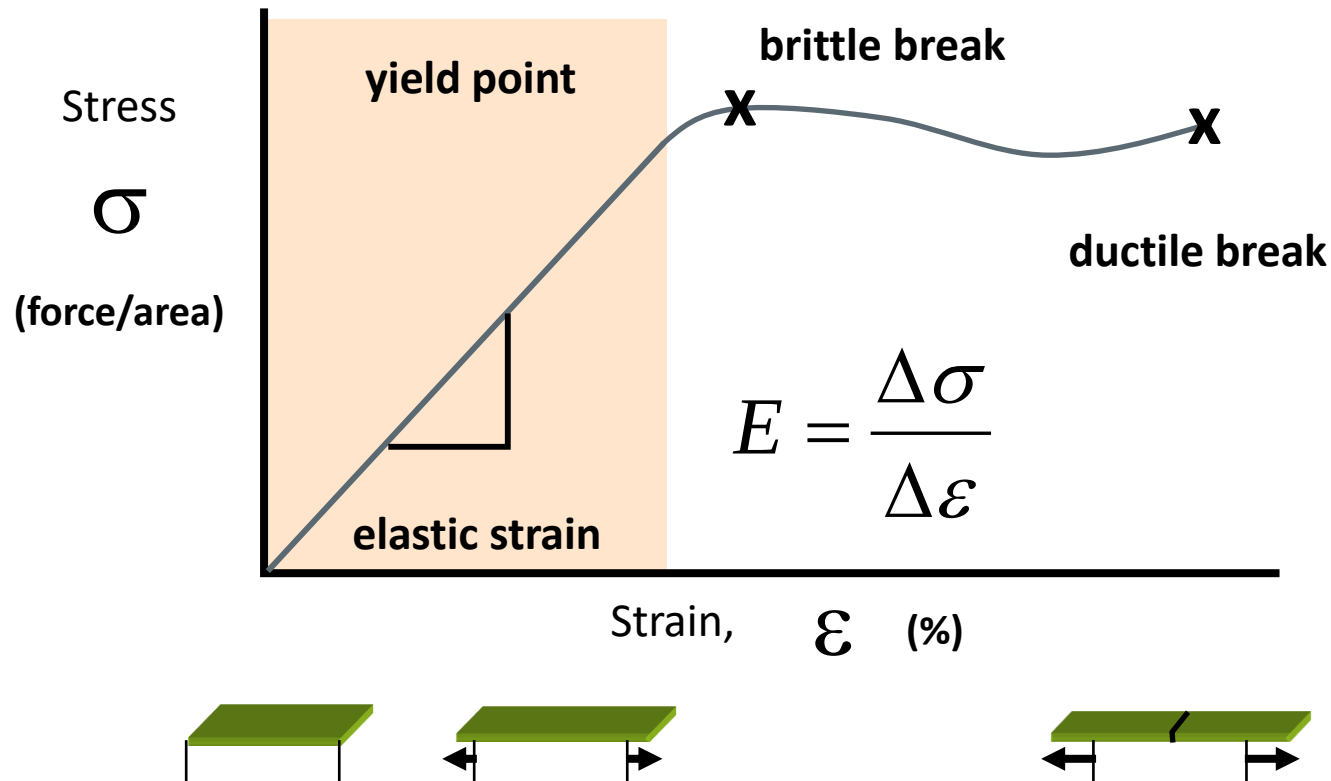
The stress (load over cross-sectional area) and percent elongation are called the *break strength* and *break elongation*.

...but the break point is uncontrolled.

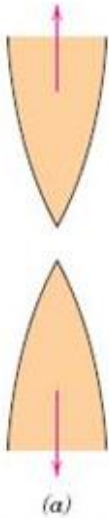


Stress Fracture Defined

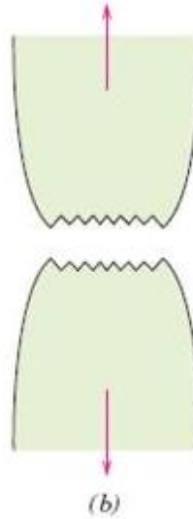
Excessive Stress or Strain will Cause a Break?



Ductile vs Brittle Fracture



Highly ductile fracture in which the specimen necks down to a point.



Moderately ductile fracture after some necking.

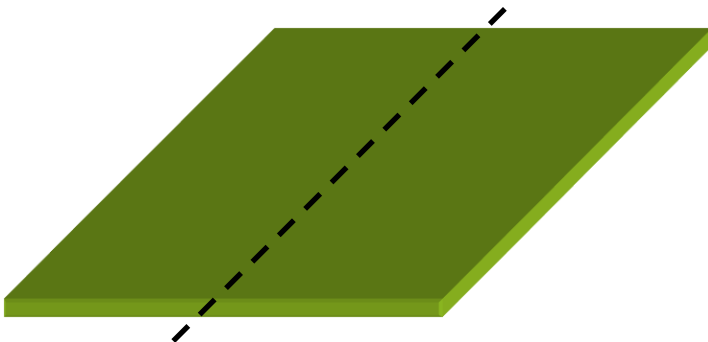


Brittle fracture without any plastic deformation.

Controlled Fracture

In slitting, we have a specific point where we want fracture to occur, so we need to control it.

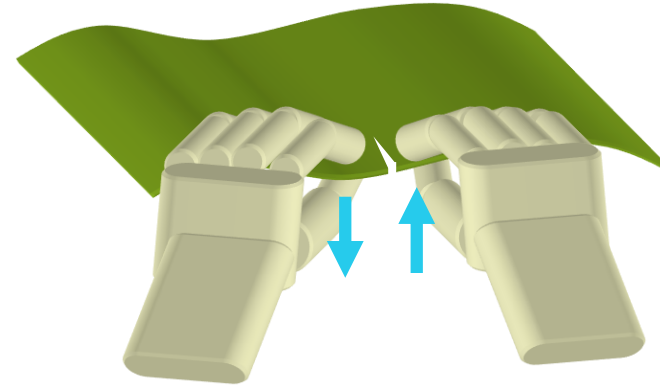
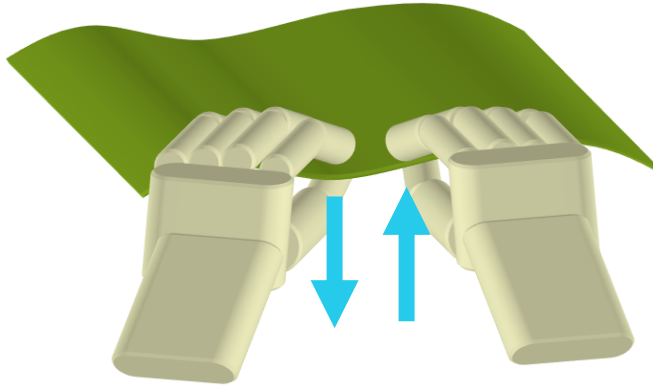
We use focused geometry to control where we create stresses high enough to fracture the material.



Fracture Force

Once the fracture is started, we want to continue it.

Many times, the crack propagation stress is lower than the crack initiation stress.



Fracture is dependent on stress (force over area) so the fracture force will increase with thickness.

Tear resistance and toughness are material properties usually dependent on fracture rate, so the fracture force may increase with speed.

Conventional Slitting Methods

All are based on fracturing the tensioned, moving web by mechanical stress concentration.

Razor

Cuts like a knife.



Crush

**Cuts like a
pizza cutter.**



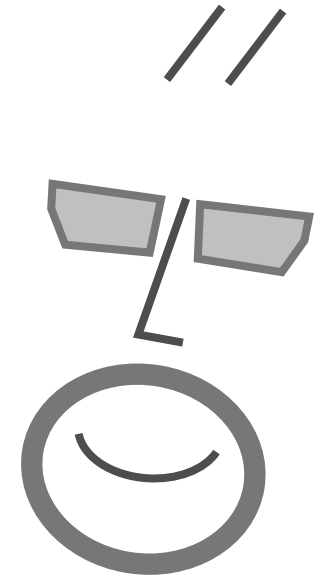
Shear

**Cuts like
scissors.**



Slitting Qs

What are the key variables of traditional slitting?



Slitting is a Process

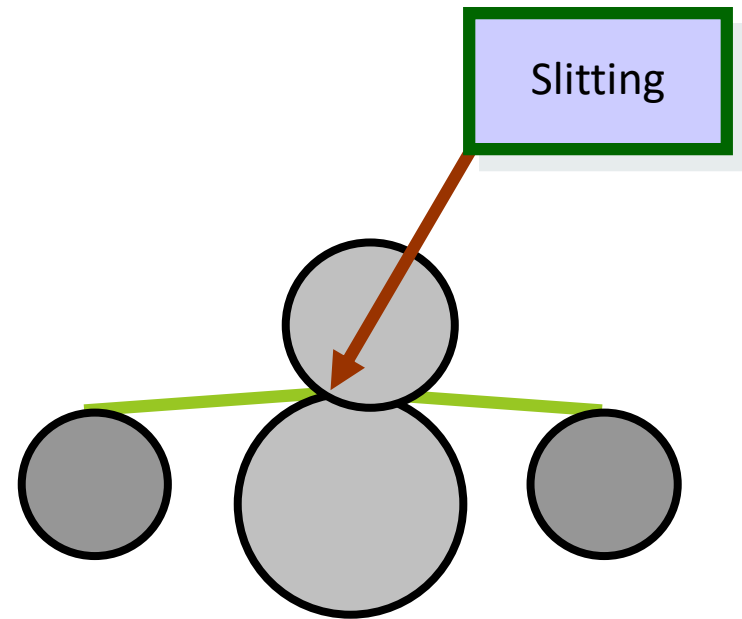
Process to control MD fracture of web products.

Slitting is not just about knives.

It's about knife-product interaction.

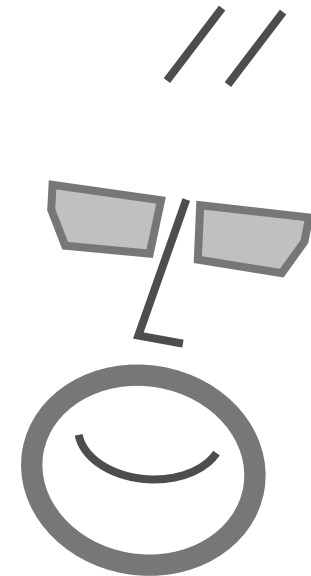
- Web Material Properties
- Knife Material and Setup
- Web (and Knife) Speed Control
- Web Tension Control
- Web-to-Knife Contact

Goal: Edge Quality
Correct Position & Width
Minimum Debris

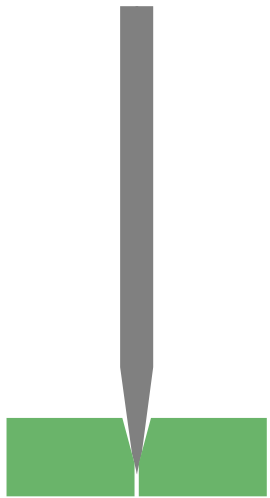


Slitting Qs

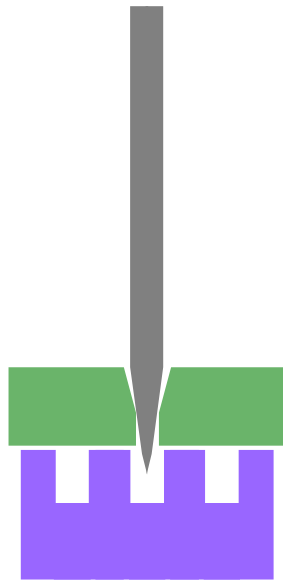
What are the traditional slitting methods?



Knife Profiles

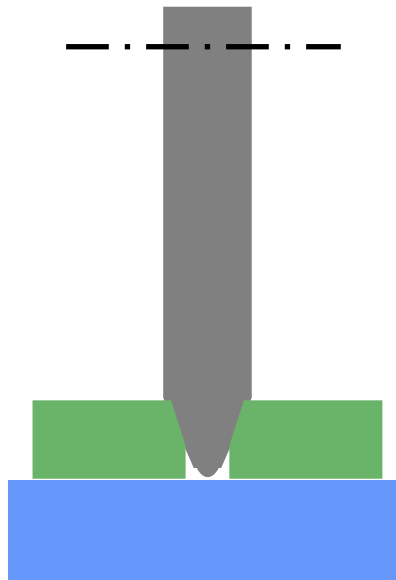


**In
Air**



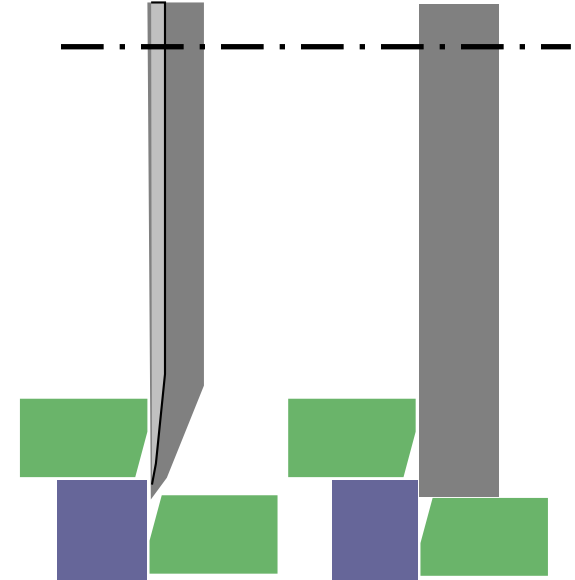
In Groove

Razor



**Crush Knife and
Anvil**

Crush



**Profile Top
Shear**

**Square
Shear**

Shear

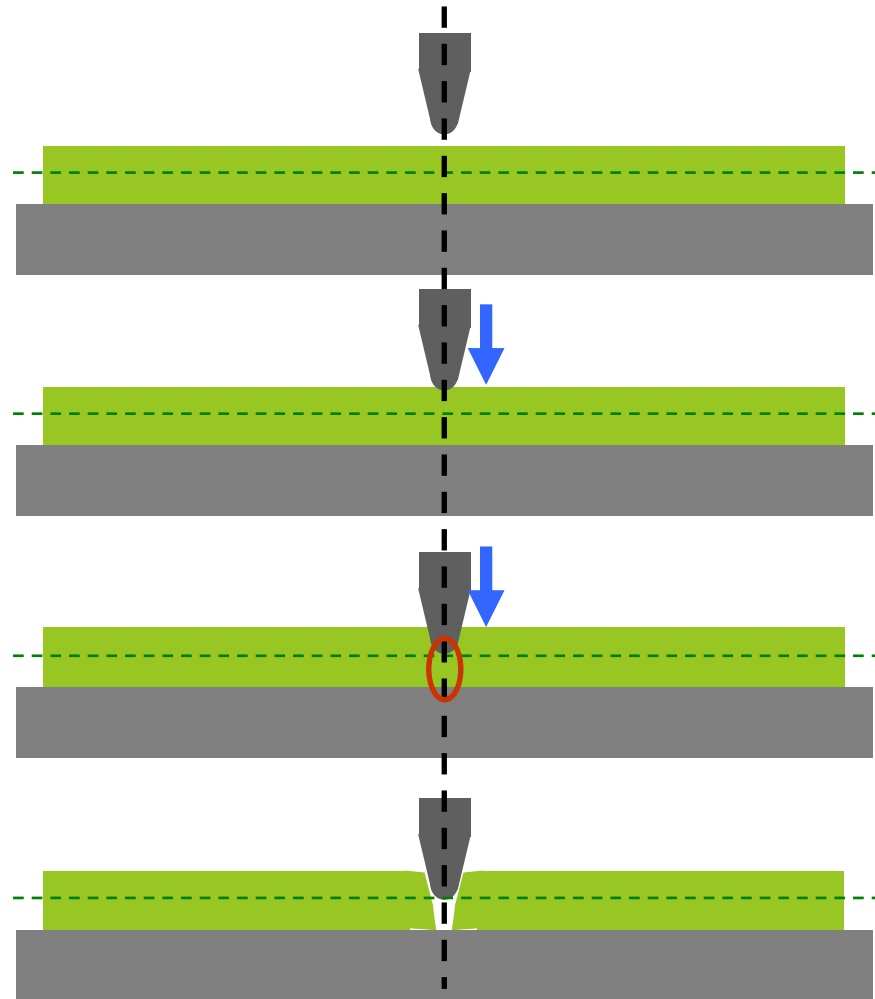
Crush Slitting = Focusing Stress

Web product with
target slit point.

Crush knife & anvil nip the
web at slit point.

Stress builds in
converging nip point.

Fracture occurs within
zone of high stress.



Two Geometries of Shear Slitting

Shear can be the most focused slitting method, concentrating stress between two sharp points.

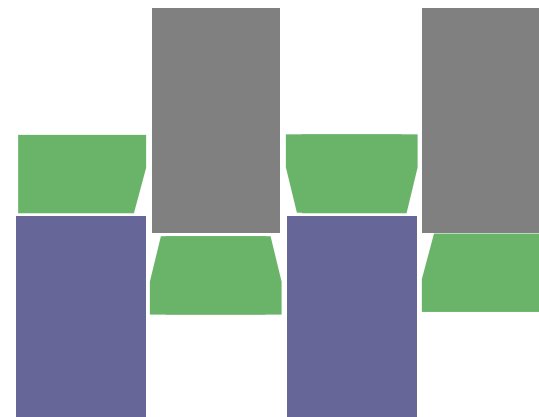
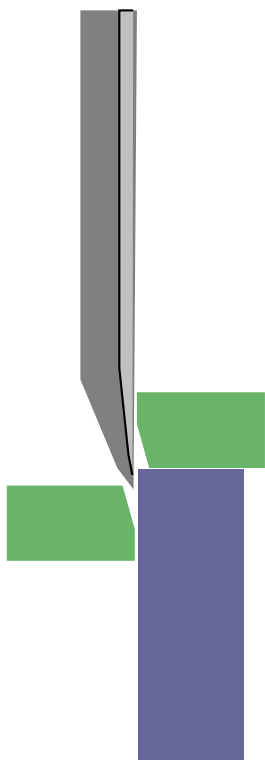
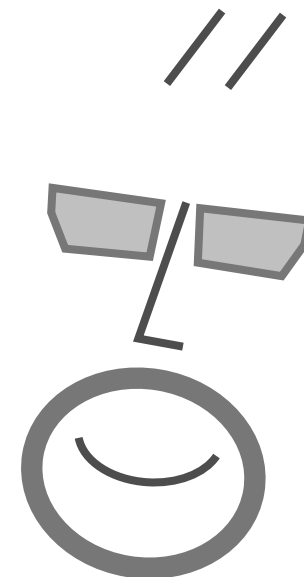


Image courtesy of Dienes Corp.

Slitting Qs

What defines slitting quality?



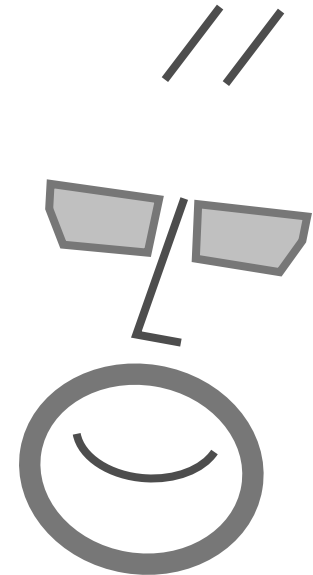
Slitting Quality

Quality Slitting =

- 1. Minimum debris / dust**
- 2. Correct slit position and width (within specifications)**
- 3. Good slit edge quality (without defects listed below)**
 - **Rough, nicked, or scratched edge**
 - **Deformed, scalloped, or wavy edge**
 - **Skiving, angel hair, or double cuts**
 - **Fusing, smearing, or delaminating coated or laminated products**

Slitting Qs

What causes slitting debris and dust?



Slitting Debris

In any mechanical slitting method, there are two sources of debris:

1. Fracture cracks that split (bifurcate), releasing particles from the body of the web
(Where does the kerf go?)
2. Abrasive wear from the freshly fractured edge rubbing on the side of the knife after the cut point.

Minimum Debris / Dust

Quality Slitting =

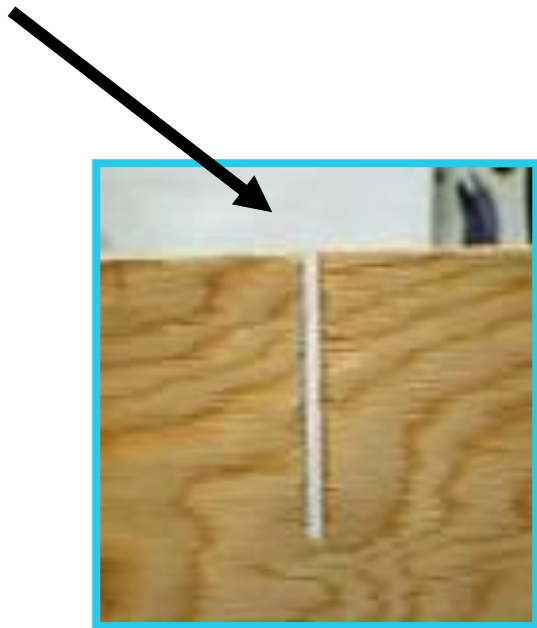
1. Minimum debris / dust

- Minimize slitting kerf by *focusing* fracture stresses through knife geometry and structural rigidity.
- Minimize edge-to-knife abrasion through optimized web path, tensioning, and knife geometry.
- Minimize knife wear. Replace or sharpen worn knives.

Debris from Slitting Kerf

Kerf is defined as the width of a groove made by a cutting tool.

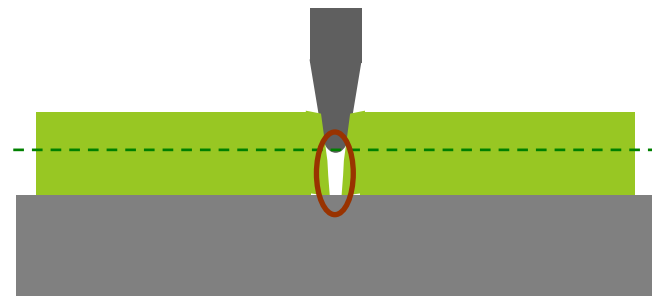
A saw blade creates an obvious kerf.



Slitting kerf is less obvious.

The entire volume of product under fracture stress levels may turn into slitting debris or dust.

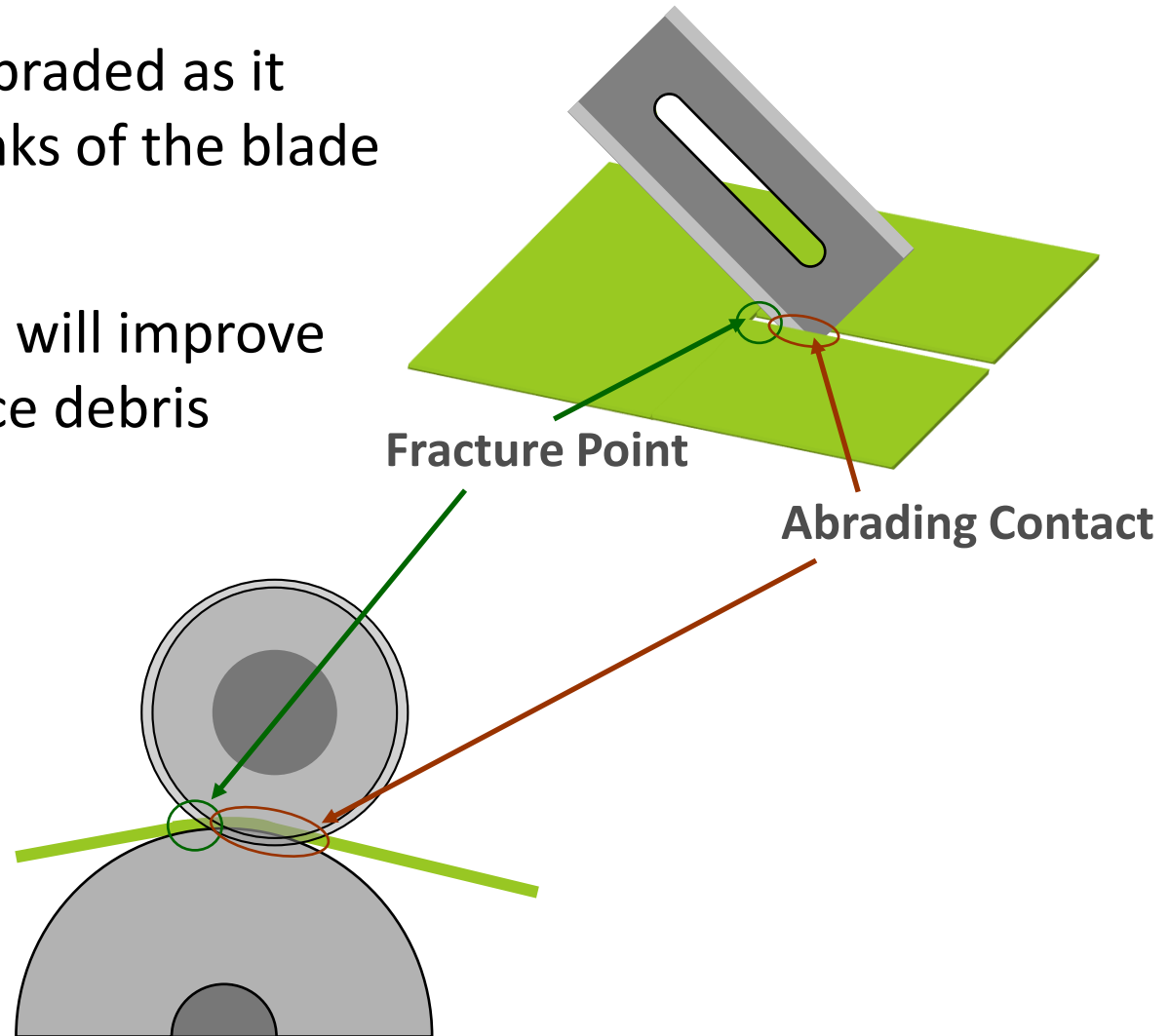
Focusing slitting geometry reduces the volume at fracture stress, related dust, ragged edges, and kerf.



Debris from Abrasion

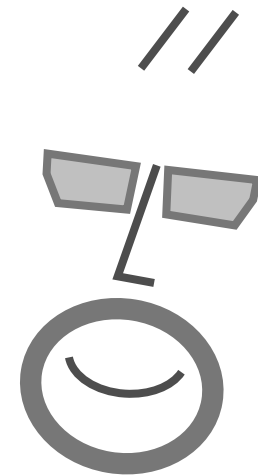
The fresh slit edge is abraded as it passed around the flanks of the blade or knives.

Reducing flank contact will improve edge quality and reduce debris generation.



Slitting Qs

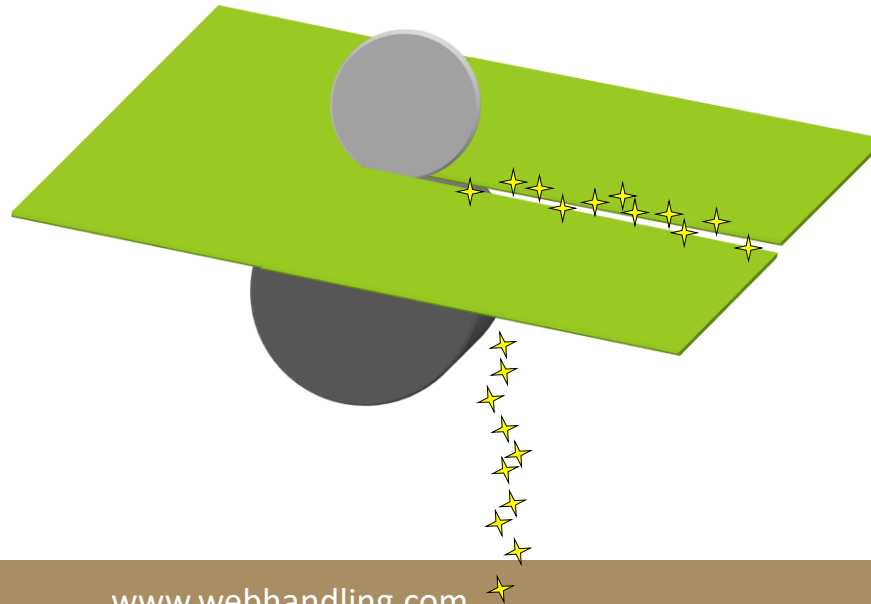
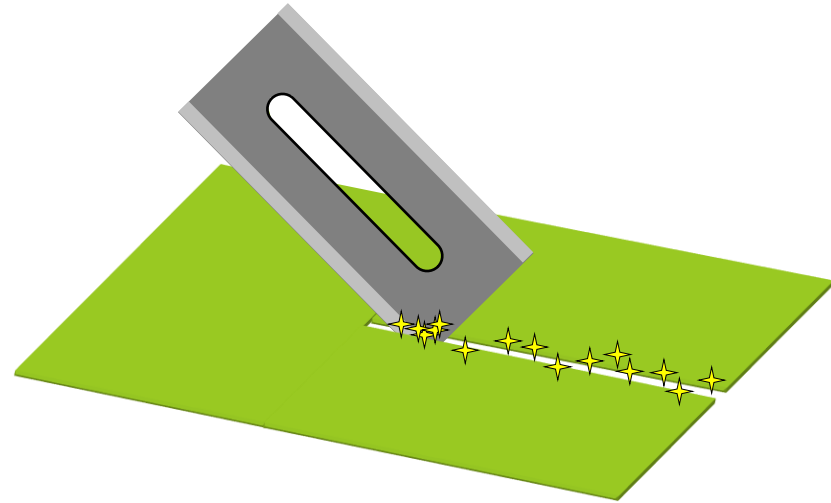
*Does slitting debris
contaminate my coated or
laminated product?*



Transport of Slitting Debris

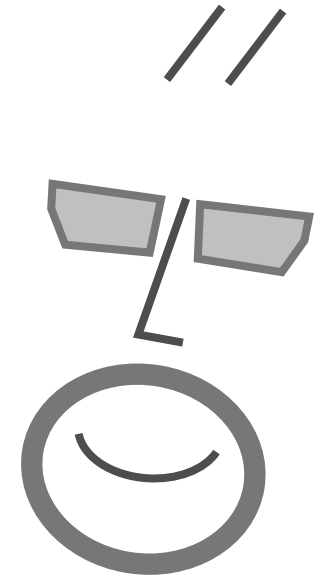
Slitting debris tends to stay near a web's edge. As debris levels increase, dust may fall off the web or become airborne.

Always avoid having the web or other debris sensitive equipment under the slitting process to avoid falling dust contamination.



Slitting Qs

What causes slit width or position variations?



Slit Width Variations

All slitting methods will have slit width variations due to the web or process control due to tension-related necking, crossweb tension variations of web bagginess, and any non-flatness or gathering in the knives.

Perfect web and tension control will still have slit width variations if the knife positions move laterally. This is largely a function of the blade and knife holder deflection and rigidity.

Razor & Crush: The blade position determines the cut point.

Shear: The bottom blade determines the cut point. Both methods can be pushed open when the product fails to reach critical fracture stress.

Correct Slit Position and Width

Quality Slitting =

2. Correct slit position and width

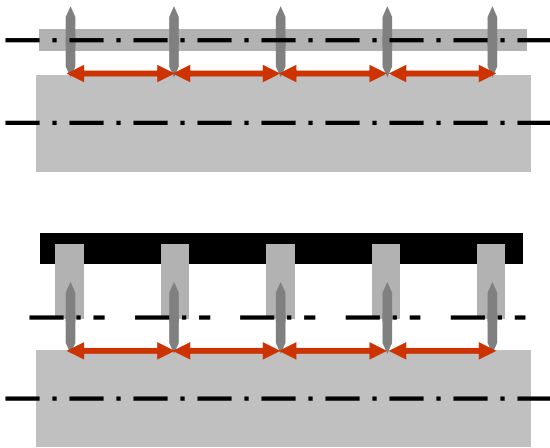
- Minimize shaft or crossbar deflection.
- Minimize blade holder deflection or slop.
- Minimize axial and radial runout.
- Minimize tension variations.
- Ensure taut, flat web at slit point.
- Set tension high enough to pull out bagginess.

Knife Positioning

Setting the slit spacing ranges from simple ruler-assisted manual positioning to sophisticated electronic-assisted manual or automatic systems.

Crush Slitting:

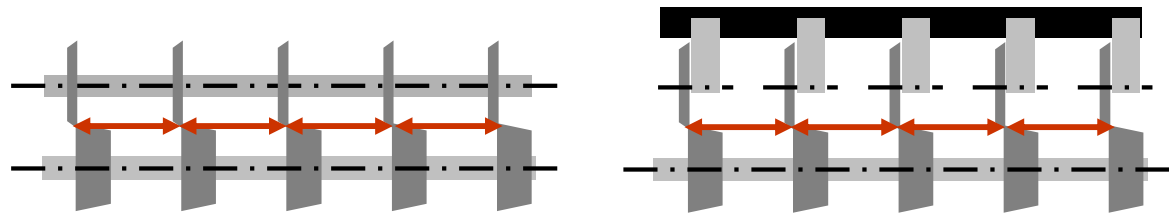
Knife-to-knife spacing determines slit spacing.



Shear Slitting:

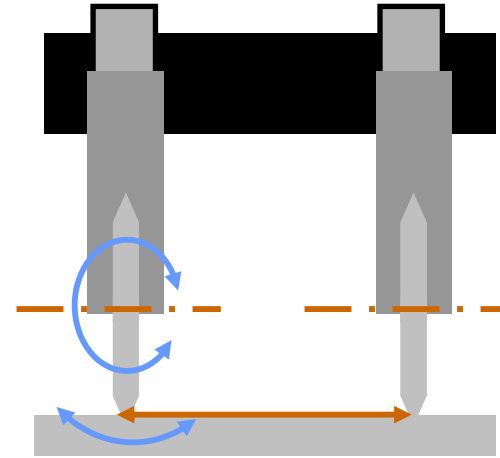
The bottom knives are typically fixed and set the slit spacing.

The top knives are loaded against the fixed bottom knives.

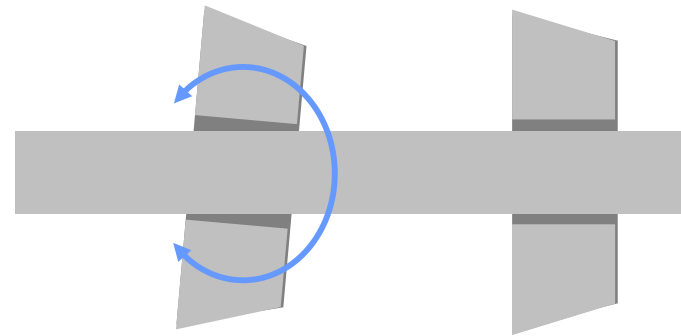


Knife Holder or Mount Rigidity

Any looseness in the rotational axis creates slit position and width variations.



Knives on common shaft may tip if there is too much clearance, creating axial runout and slit position / width variations.



Slitting Failure

Mechanical slitting is always about focusing force over a small area to reach a materials fracture stress. If stresses of below the fracture criteria, the product will fail to slit. Low stresses are cause by:

- 1) Insufficient applied force, or more commonly
- 2) Insufficient sharpness to focus the force into a high stress.

Razor: Low tension, web bagginess, and web flutter lead to insufficient applied force and failure to cut

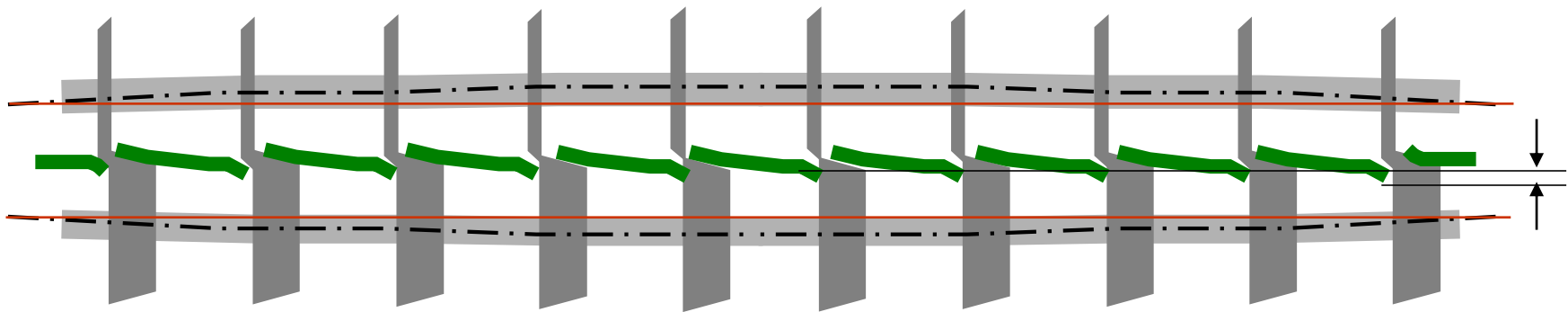
Score & Shear: Both methods can be pushed open when the product fails to reach critical fracture stress.

(Fibrous products are especially susceptible to failure to slit.)

Deflection from Slitting Forces

The slitting force will create deflection in both the top and bottom shafts. Knife engagement must be greater than deflection .

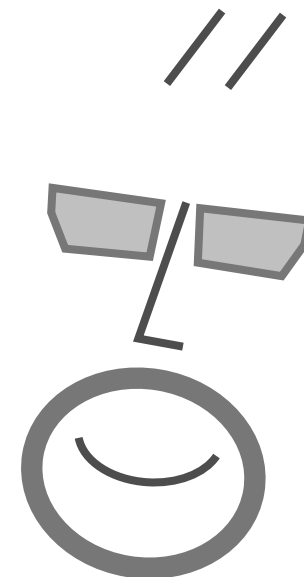
Deflection creates edge to center slitting differences.



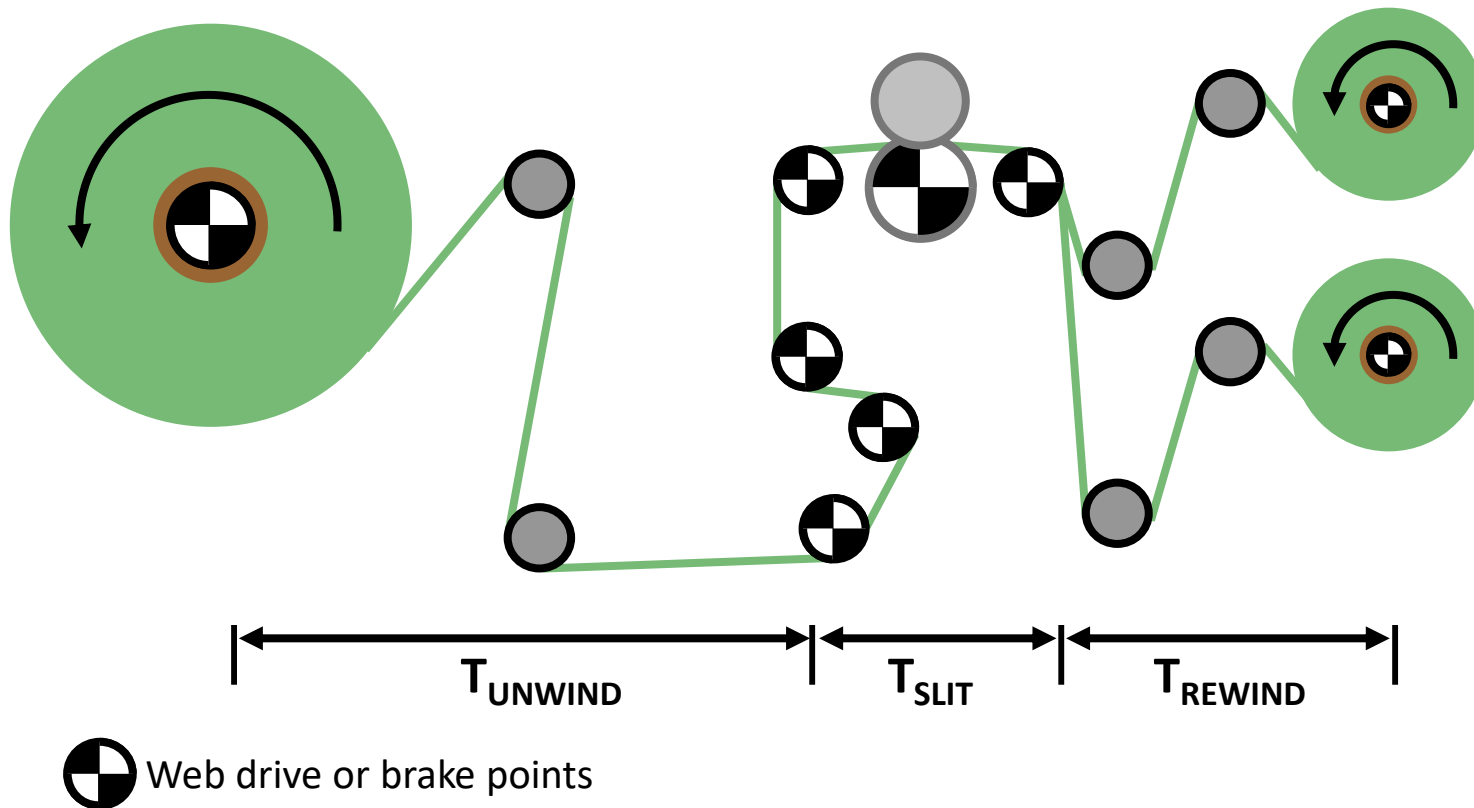
With more knives per width, narrower slitting
will have more deflection

Slitting Qs

How is tension typically controlled at slitting?



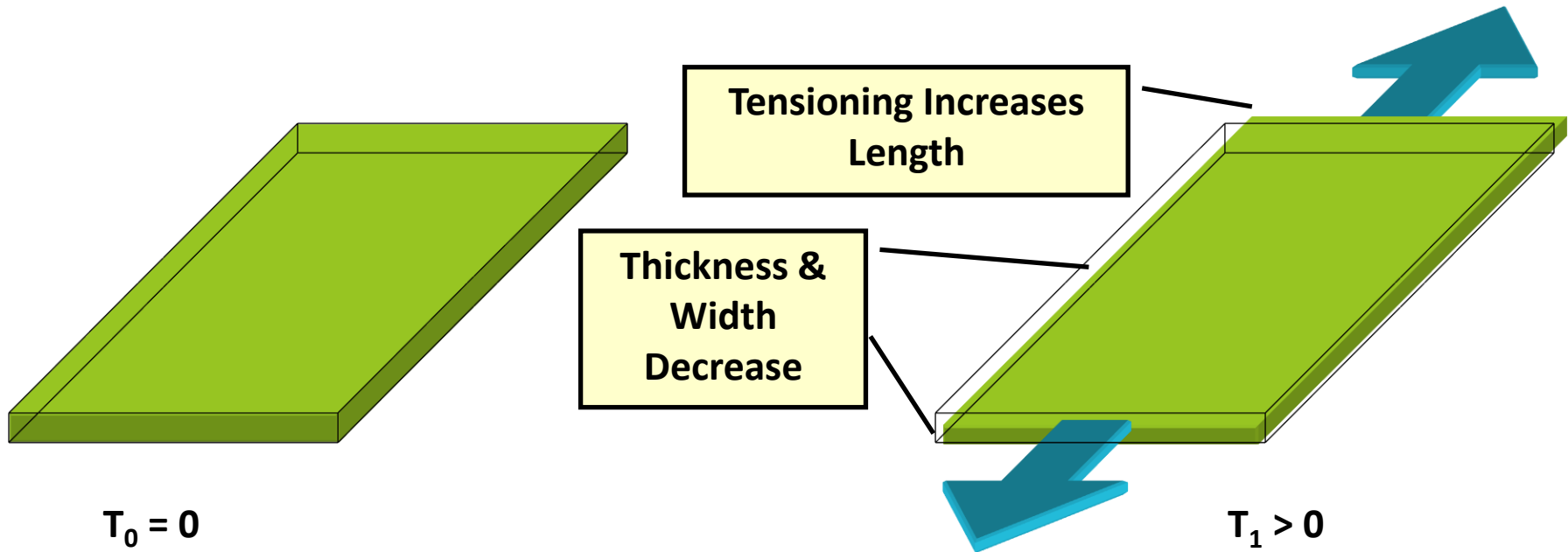
Typical Slitter Rewinder Tension Control



Tension, Strain, & Necking

For solid materials, tension and compression stresses do not significantly change density.

Therefore, dimensional increases in one direction are offset by decreases in other directions.

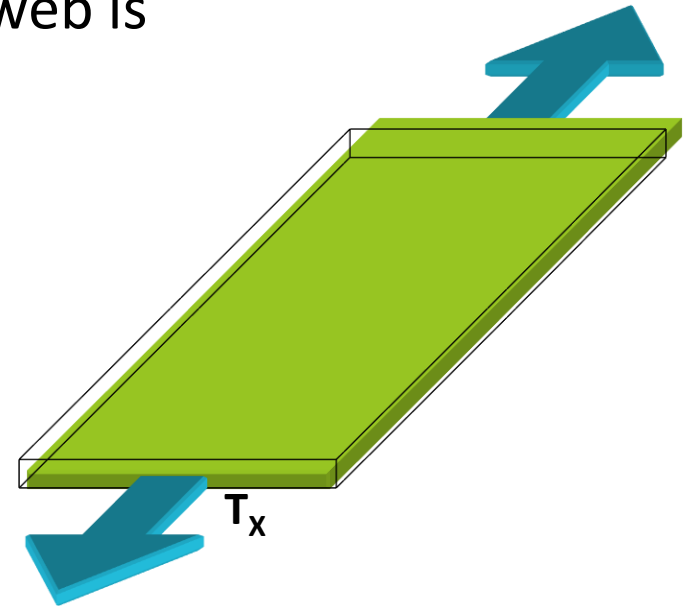


Necking and Slit Width

Tension, thickness, width, modulus (stretchiness) and Poisson's ratio determine how much the web is necked at slitting.

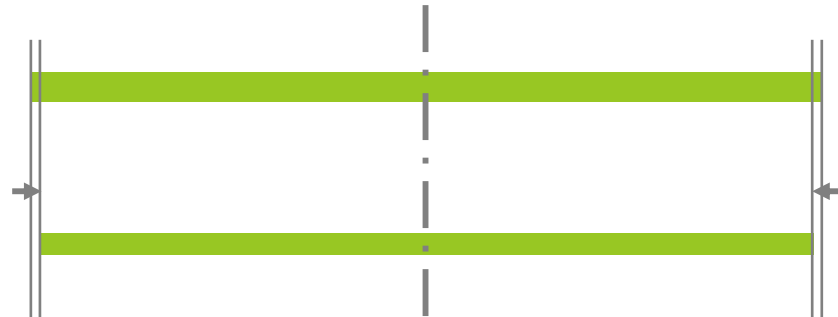
$$\varepsilon_Y = -\nu\varepsilon_X = -\nu\left(\frac{T_X}{twE}\right)$$

$$\delta w_T = w_0(-\nu\varepsilon_X)$$



No Tension

Tensioned

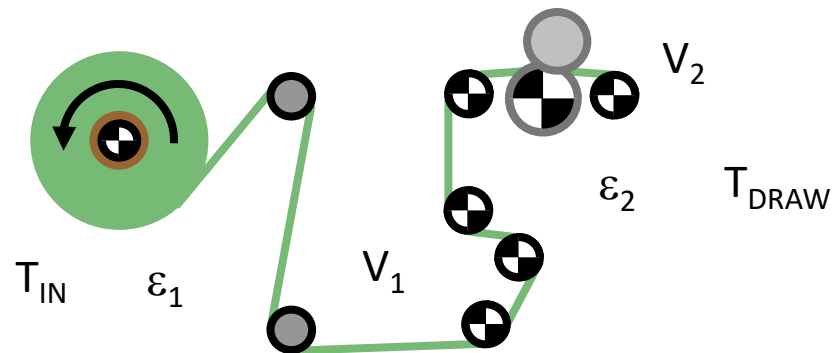
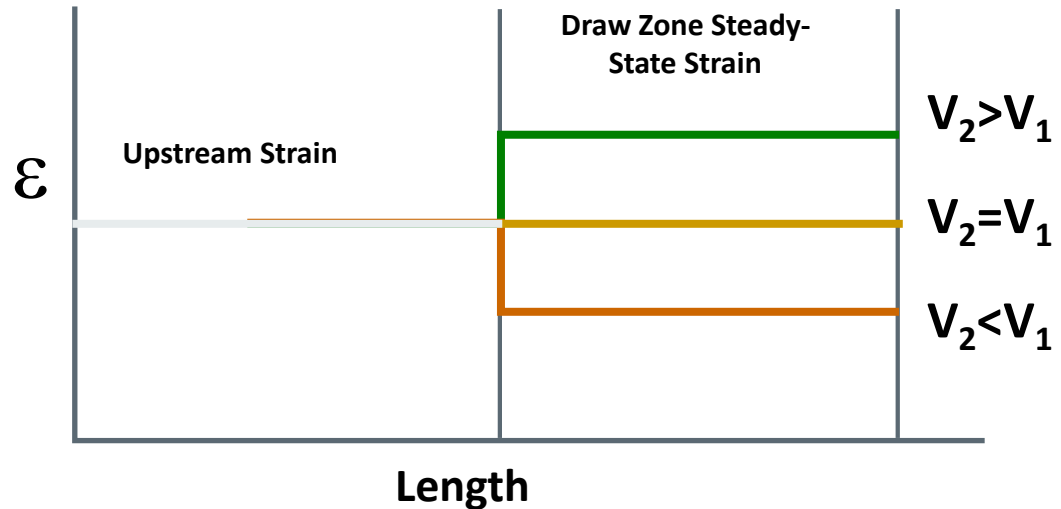


Draw Strain and Velocities

Draw zone strain alters the incoming web's strain based on roller surface velocity changes.

A velocity increase will increase draw zone strain and tension.

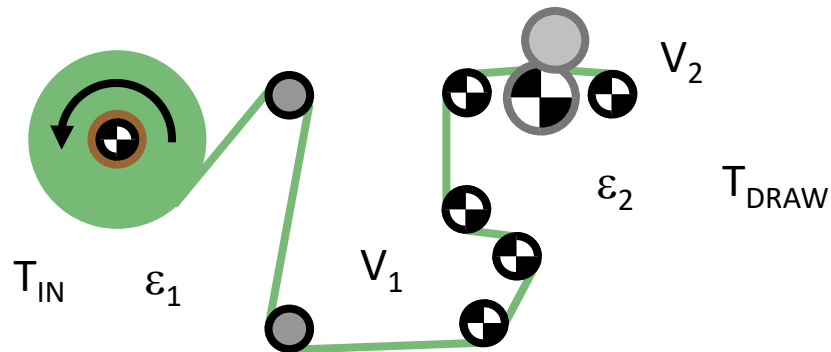
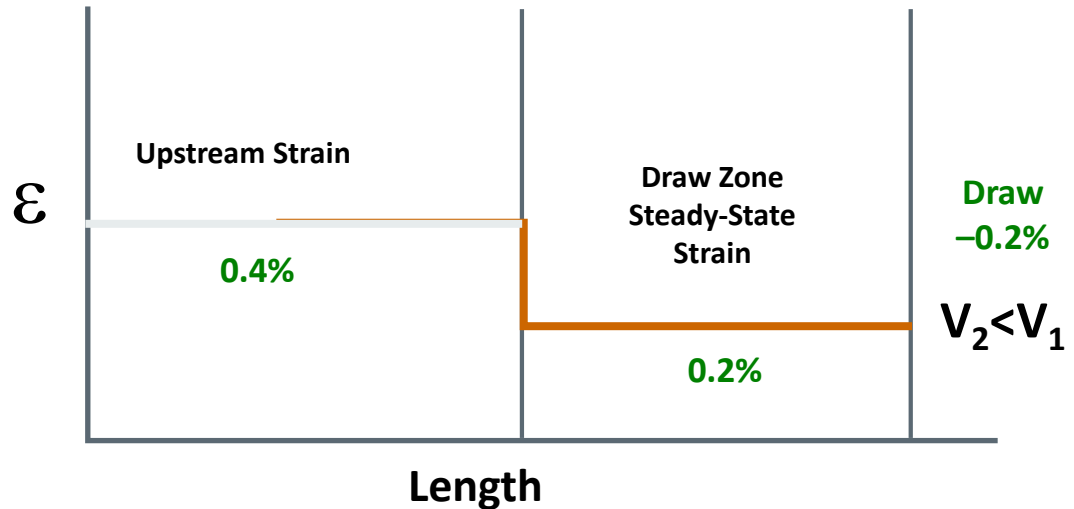
A velocity decrease will decrease draw zone strain and tension.
(Too great a drop will make the web go slack.)



Negative Draw

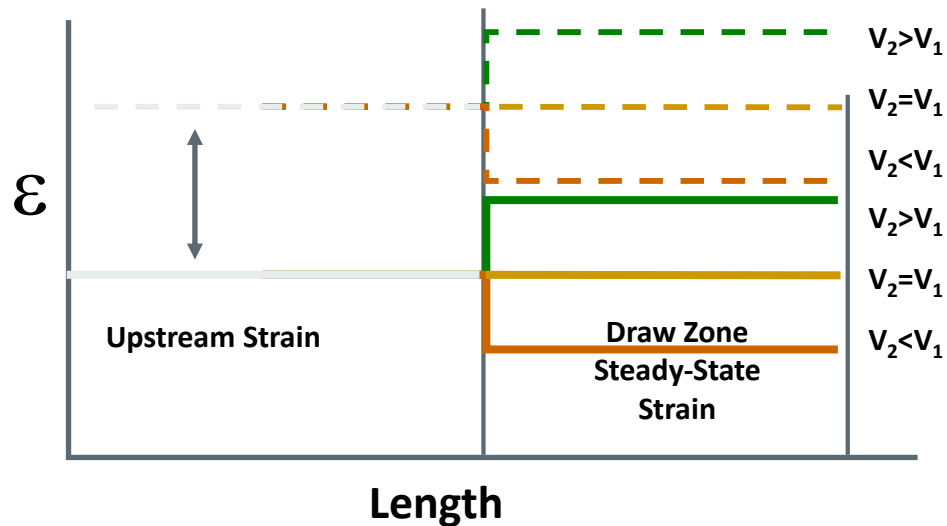
Negative draw, where the downstream velocity is less than the input velocity, is often believed to create slack web.

However, if the percent speed drop is less than the input web strain, the web will NOT go slack.

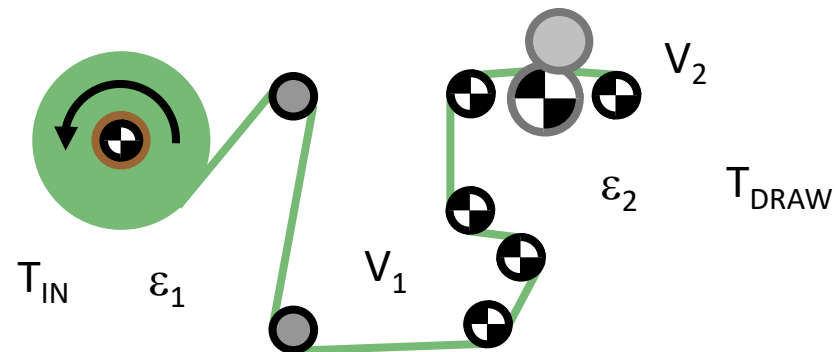


Draw Strain and Input Strain

Draw zone strain is always relative to the upstream strain.

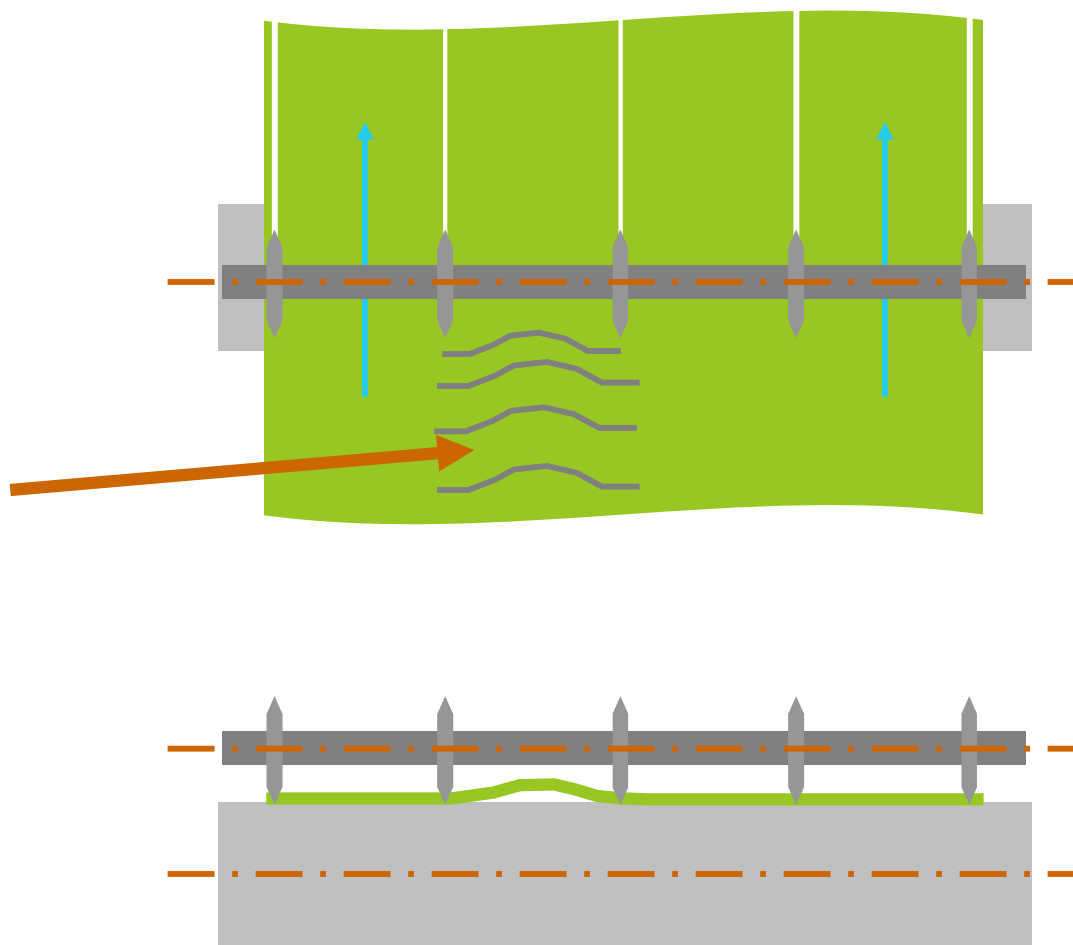


Changes in upstream tension or strain will feed into and change the draw zone tension and strain.



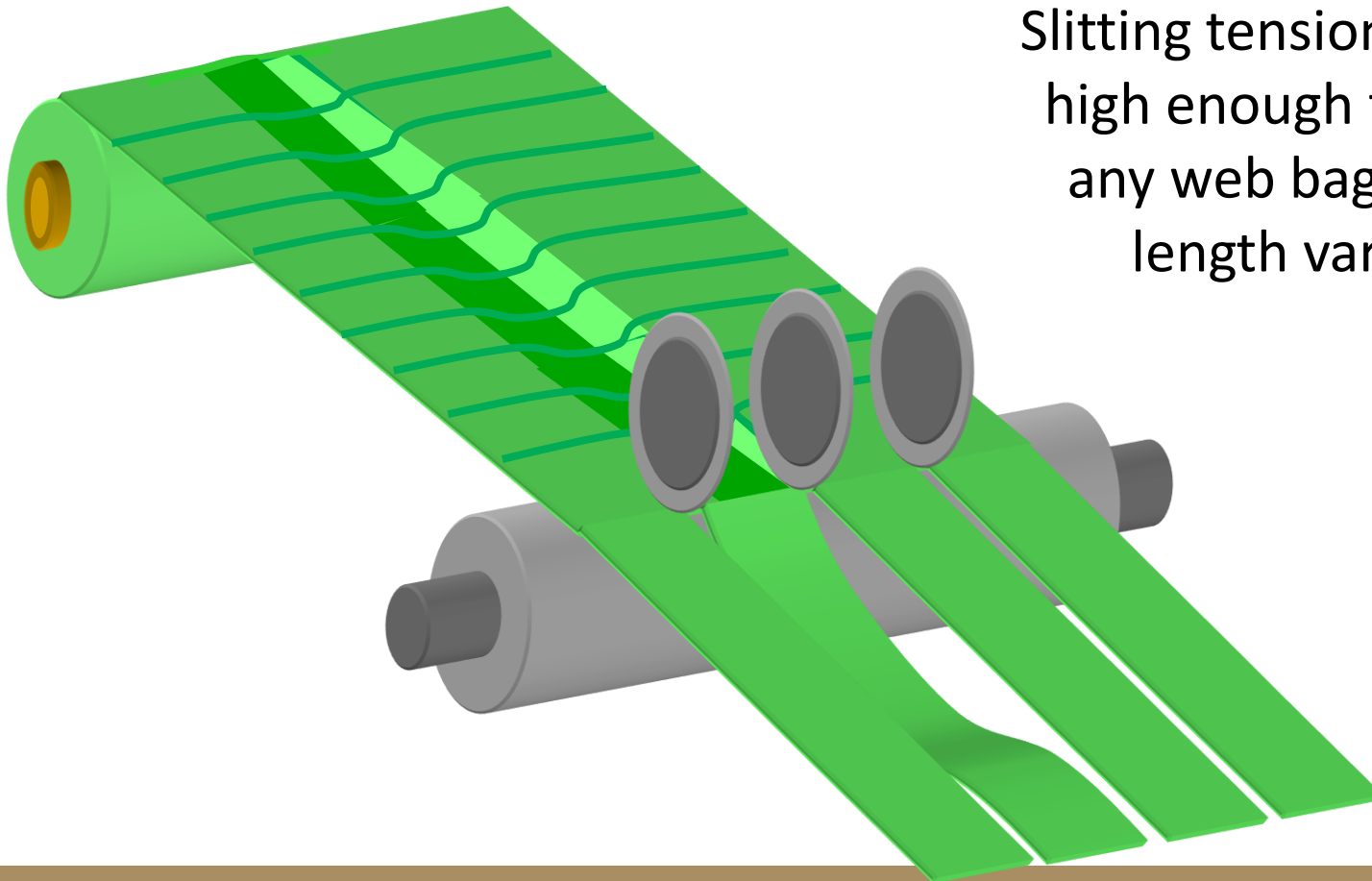
Pre-Slit Web Flatness

Any lateral buckling entering the slitting knives will result in a greater post-slit, untensioned strand width.



Tension Pulls Out Bagginess

Loose web will have a poor slit edge quality.



Slitting tension should be high enough to pull out any web bagginess or length variation.

Slitting Qs

How does razor slitting work?

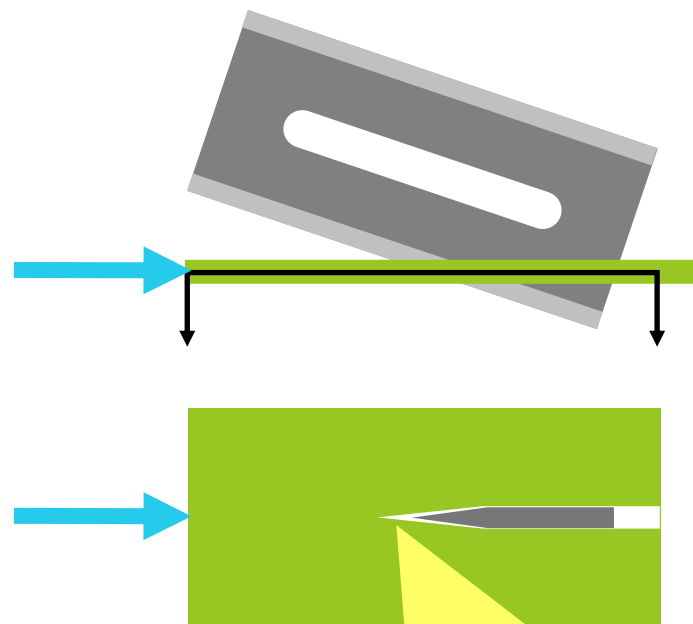


Crack Propagation

Web tension focused onto a small area edge of a razor blade = high stress and fracture.

Once a crack is initiated, slitting fracture and separation may occur ahead of the web-knife contact point.

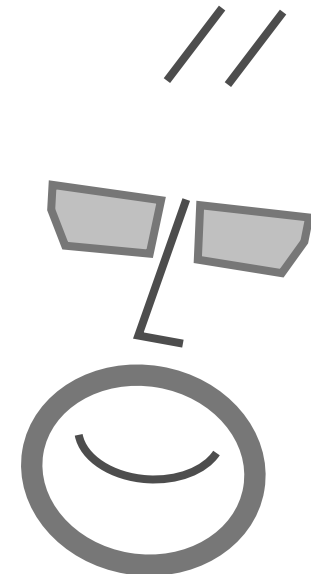
The knife wedge causes the existing crack to move forward.



The often crack propagates ahead of the knife tip like ice in front of an icebreaker ship.

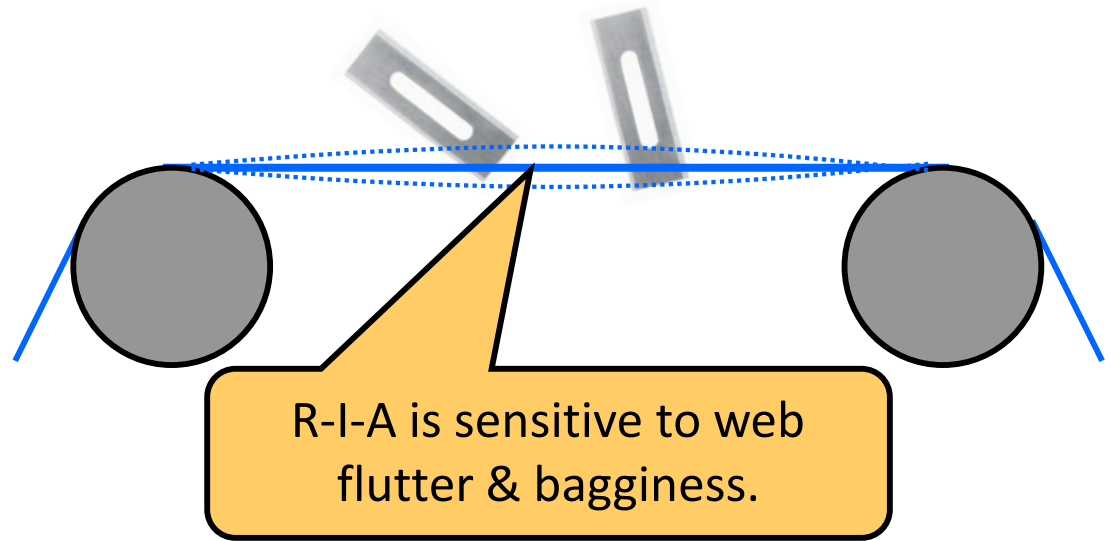
Slitting Qs

How is razor slitting optimized?

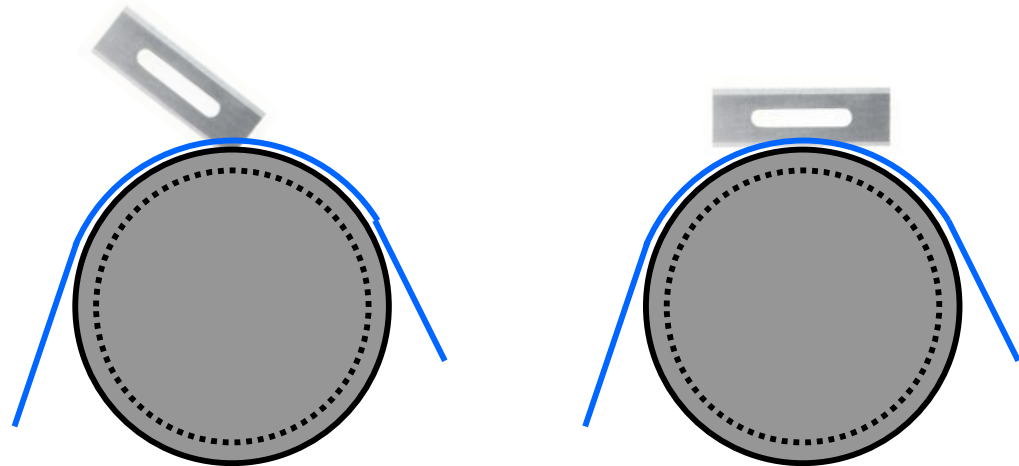


Razor Slitting

Razor-In-Air



Razor-In-Groove



Razor Slitting = Focusing Stress

Razor-In-Air

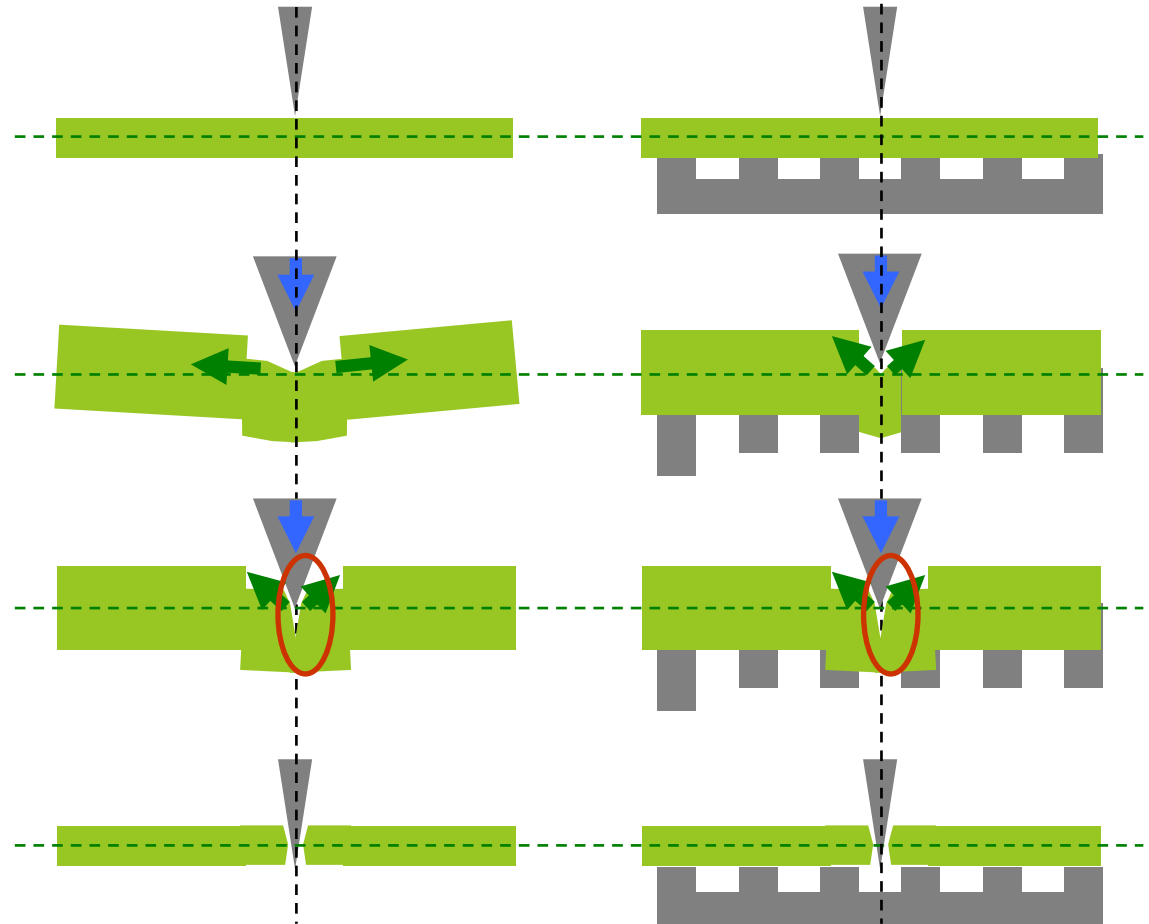
Razor-In-Groove

Web product
with target slit
point.

Razor blade begins
to enter the web path.

Web stiffness and
tension creates stress
near razor tip.

Fracture occurs within
zone of high stress.



Razor-In-Air vs. Razor-In-Groove

Razor-In-Air



Razor-In-Groove



Application Factors

Easiest installation

Most width flexible

May create wavy edge in ductile webs

Easy to oscillate

Sensitive to flutter & bagginess

Needs safety guard

Requires grooved roller

Discrete widths

Minimizes wavy edge in ductile webs

Difficult to oscillate

Less sensitive to flutter & bagginess

Safer, roller helps guard

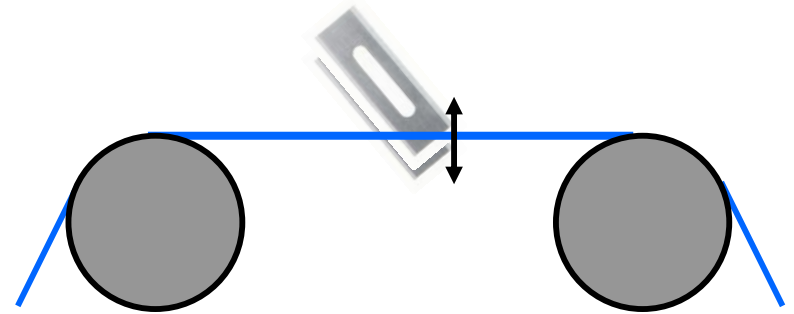
Products

Thin, non-extensible films, foils, laminates

Thin extensible films and laminates

Extending Razor Blade Life

- Oscillate blades to expand the wear zone.
- Hard coat or ceramic blades.
- Avoid overheating blades (reduce speed or product thickness)



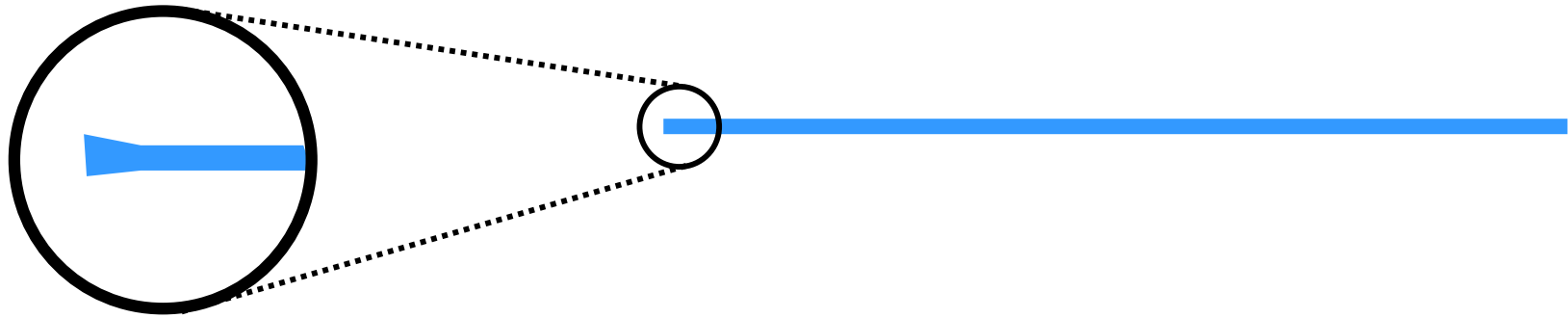
Examples: Razor Blade Holders



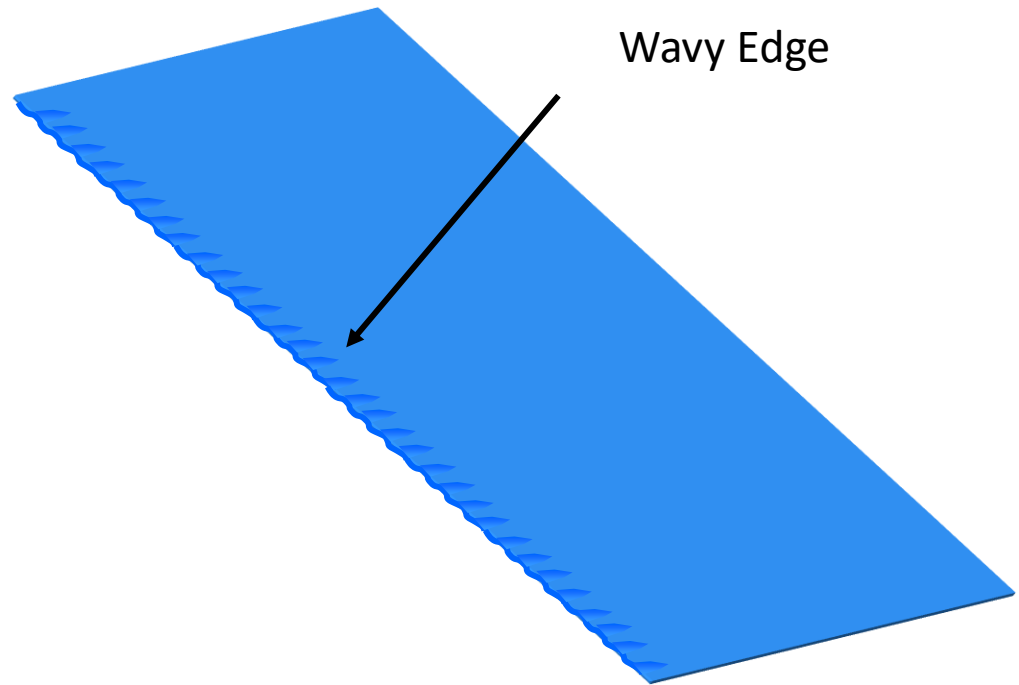
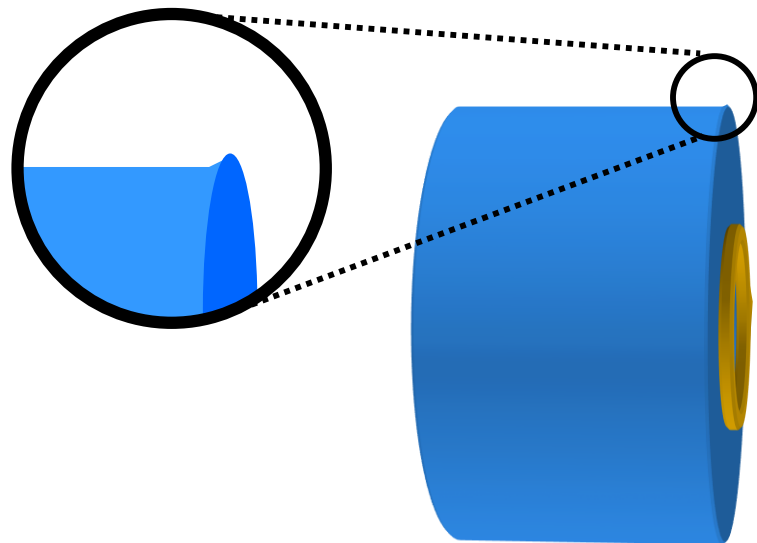
Auto-Positioning Razor Slitting



Raised Edge, Wavy Edge



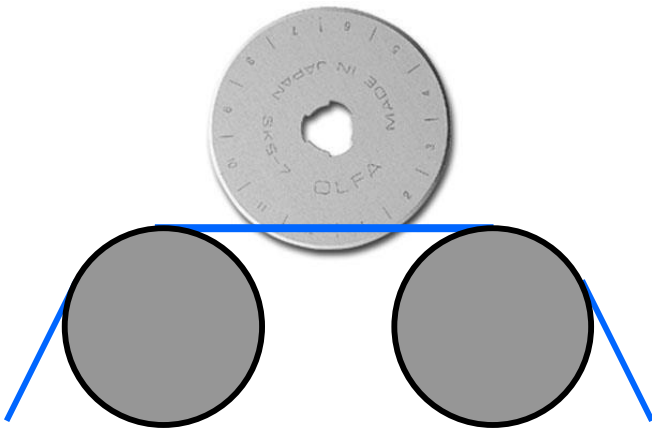
Raised Edge



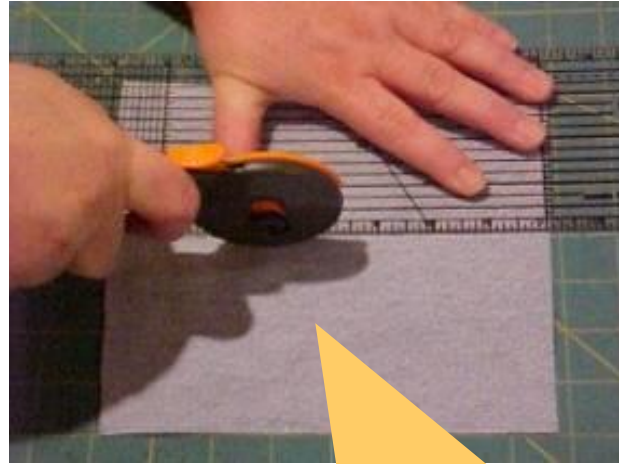
Wavy Edge

Razor Slitting

Circular Razor



Yikes! These are scary safety-wise.

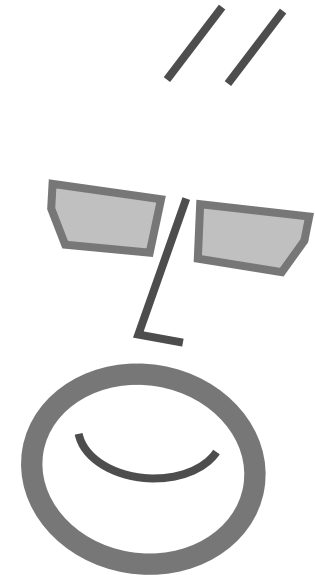


Rotary cutters are commonly used by quilters to cut fabric. The circular razor blades are used similar to crush knives, but against a self-healing mat rather than an anvil.



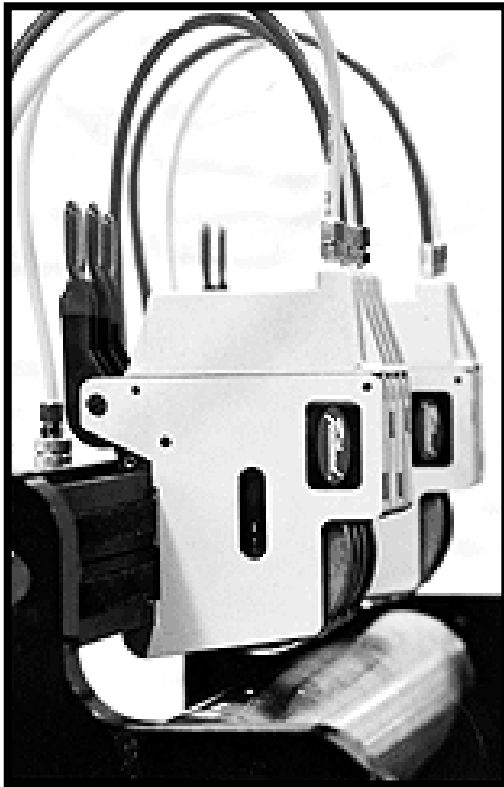
Slitting Qs

How does crush slitting work?



Crush and Shear Dovetailed Holders

Crush knives with individual dovetail holders.



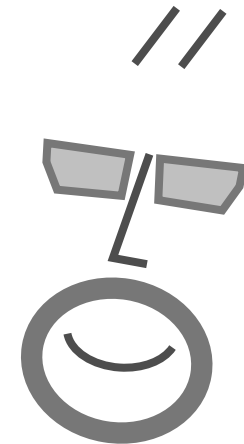
Knife holders for increasingly demanding applications.



Images courtesy of Tidland Corp.

Slitting Qs

What is the difference between crush slitting and rotary die cutting?



Two Approaches to Crush Slitting

Individual pneumatically loaded circular blades and common anvil roller

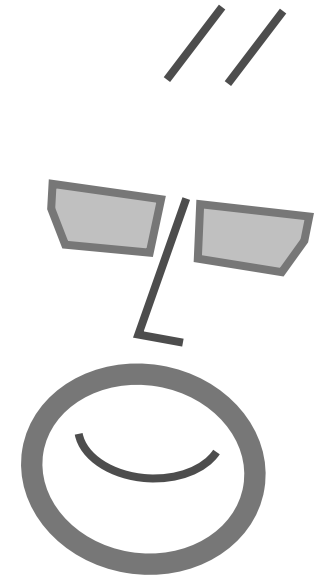


Rotary converting die: rigid or flexible tooling in tight tolerance gap vs. anvil roller



Slitting Qs

How does shear slitting work?

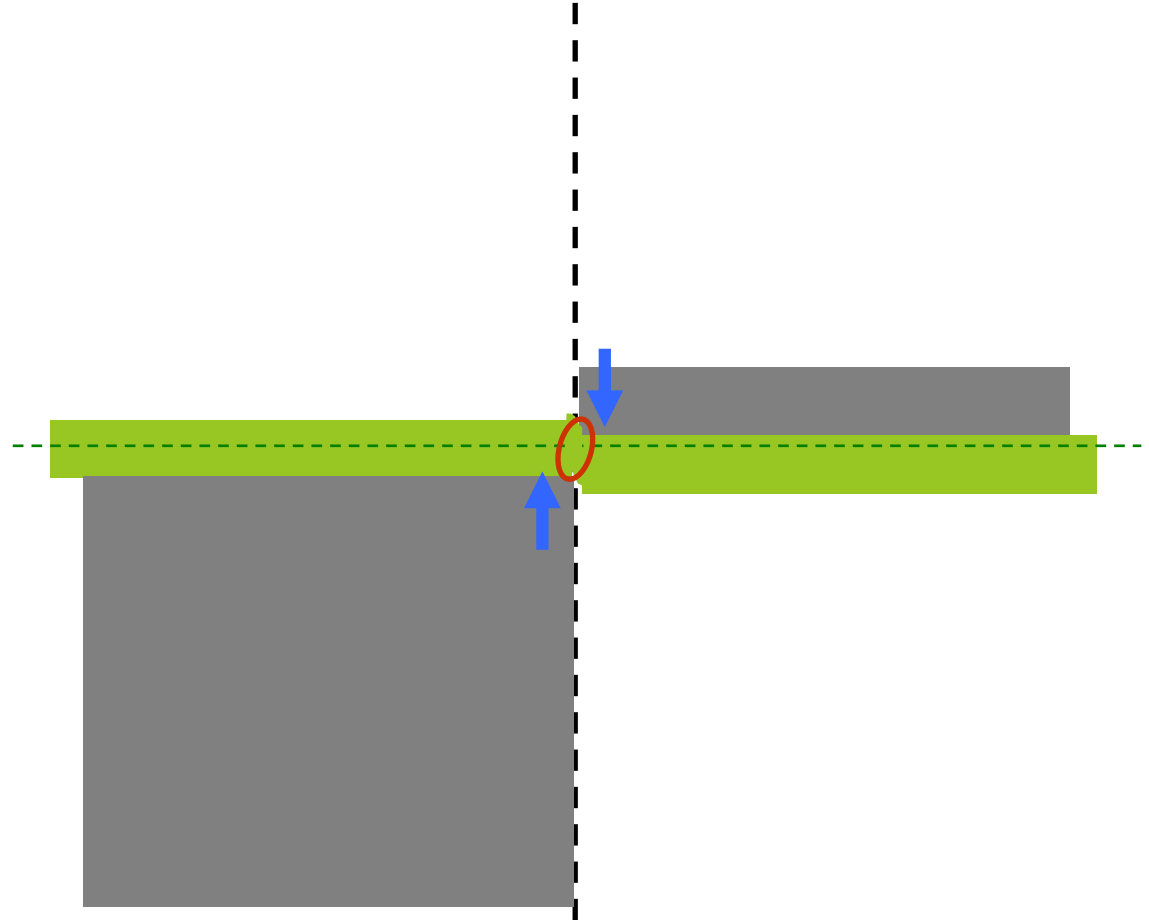


Shear Slitting = Focusing Stress

Web product with
target slit point.

Shear knives nip the
web at slit point.

**Stress builds in
converging nip point.**



Shear Slitting = Focusing Stress

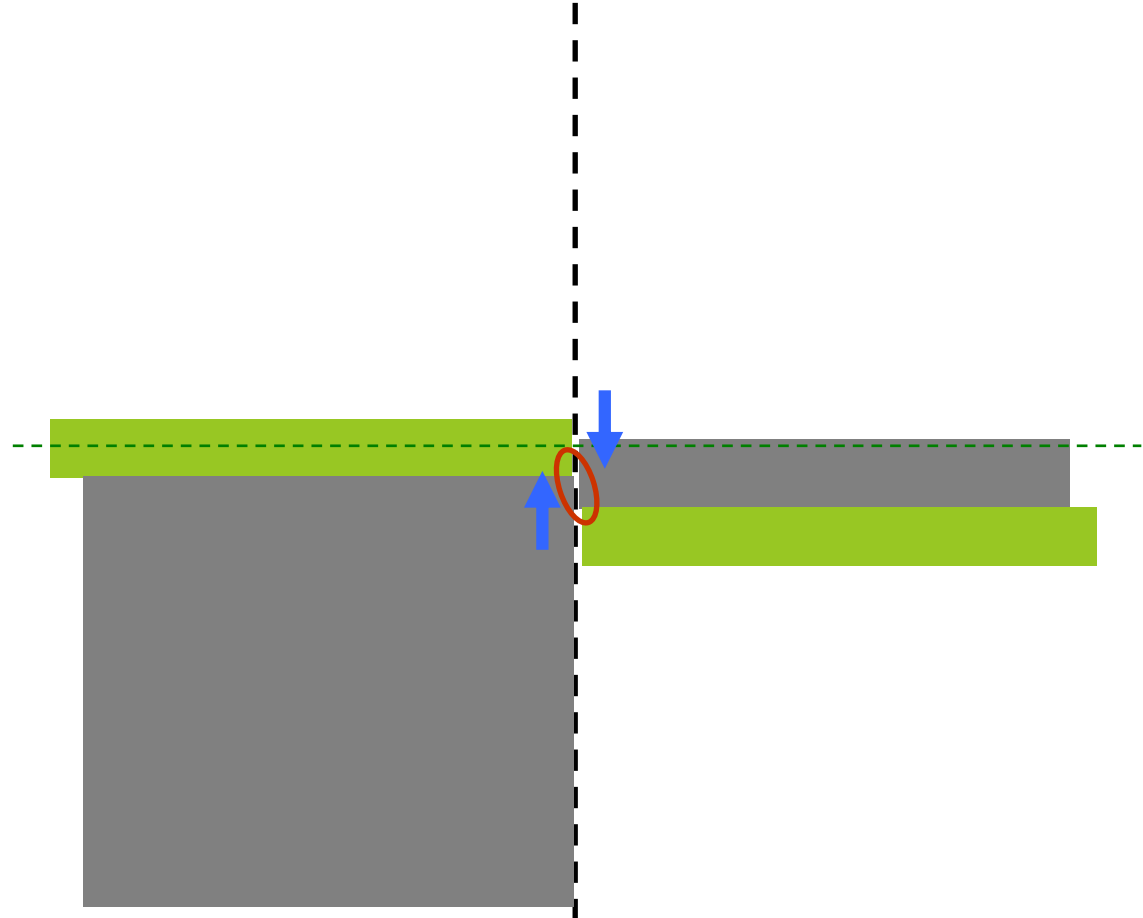
Web product with
target slit point.

Shear knives nip the
web at slit point.

Stress builds in
converging nip point.

Fracture occurs within
zone of high stress.

Post-fracture abrasion of
edge (and knives).



Shear Slitting Optimization

- ▶ Overlap
- ▶ Runout: Axial and Radial
- ▶ Angles of Shear
 - ▶ Roller Wrap Angle: “Wrap” vs. Tangential
 - ▶ Top Knife Bevels
 - ▶ Cant
 - ▶ Rake
- ▶ Edges: Sharpness and Nicks
- ▶ Surfaces

Shear Overlap and Runouts

Overlap

- ▶ Greater overlap affects the post-fracture abrasion zone, increasing the contact length of abrasion and the lateral force in the abrasion zone (since the web has a wide point on the top knife to pass by).
- ▶ Greater overlap will create a greater speed differential in undriven top blades.

Runout, Axial and Radial

- ▶ Excessive runout (over 2-5 mils) in either axial or radial directions will increase debris generation.
- ▶ Greater radial runout requires increase knife overlap to prevent top knives from jumping the bottom knives.
- ▶ Both radial and axial runout will shift the fracture focal point, leading to lack of focus as the fracture zone shifts position.

Wrap vs. Tangential Shear Slitting

- ▶ Wrap shear will have a more stable focal point, less sensitive to web bagginess or flutter.
- ▶ Wrap shear will be more difficult for the web cut on the top knife side to deflect under the blade since the web will be stiffened by the shaping around the bottom knife.
- ▶ This makes it more likely the web will not deflect under the top knife, instead buckling laterally and rubbing on the side of the top knife, likely leading to more abrasion debris generation.
- ▶ In tangential shear, without the shape stiffening of the bottom knife wrap, the web will have less difficulty bending down under the top knife.
- ▶ Tangential shear will more likely benefit from the extra wide, low small blade angle top knives.

Top Knife Bevel Angles

- ▶ There are three bevel angles and a bevel length that define the difference among top knives, in addition to their blade width.
- ▶ The first bevel angle defines the general taper of the blade from squareness.
- ▶ The second bevel angle changes the taper of the blade in the final angle to the blade tip.
- ▶ Both the first and second bevel angles define the shape of the top knife away from the bottom knife.
- ▶ A double bevel blade has bevels on both sides of the knife.
- ▶ Unlike razors and other standard blades, a top shear knife will sometimes have a 'hollow ground' (a.k.a. undercut or back bevel) on the side contacting the bottom knife.

Cant and Rake Angles

Cant Angle

- ▶ The cant angle ensures the shear knives close at the leading edge of the overlap zone.
- ▶ The cant angle doesn't need to be much, usually less than one degree.
- ▶ If the angle is reversed, opening the cut point, the fracture zone will grow, creating more debris, a poor cut edge, or fail to cut entirely.

Rake Angle

- ▶ The rake angle combined with back bevel promotes contact at the leading (and trailing) point of the overlap zone.
- ▶ A rake angle will tend to naturally form in the loading of the top blade as the side force is applied at the blade's center point, but opposed at the blade's tip.

Edges and Surfaces

Blade Sharpness and Nicks

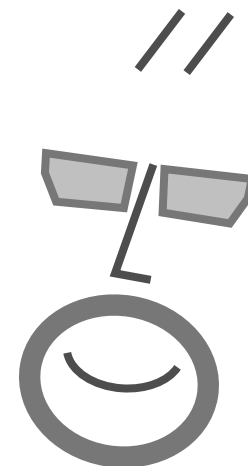
- ▶ Blade sharpness is one of biggest factors in volume of material reaching fracture stress.
- ▶ Dull blades will apply the available load over a greater volume, requiring more force to cut and creating more debris.
- ▶ Overly sharp blades are more prone to tip damage.
- ▶ A nicked or damaged blade edge will create an upset to the fracture focal point and more debris in the focal transitions.

Surfaces

- ▶ Abrasive wear of shear slitting can be reduced with highly polished blade sides or Teflon coating.

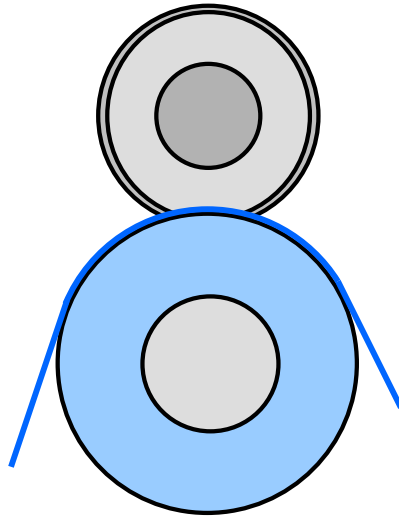
Slitting Qs

What's the difference between wrap and tangential shear slitting?



Wrap vs. Tangential Shear Slitting

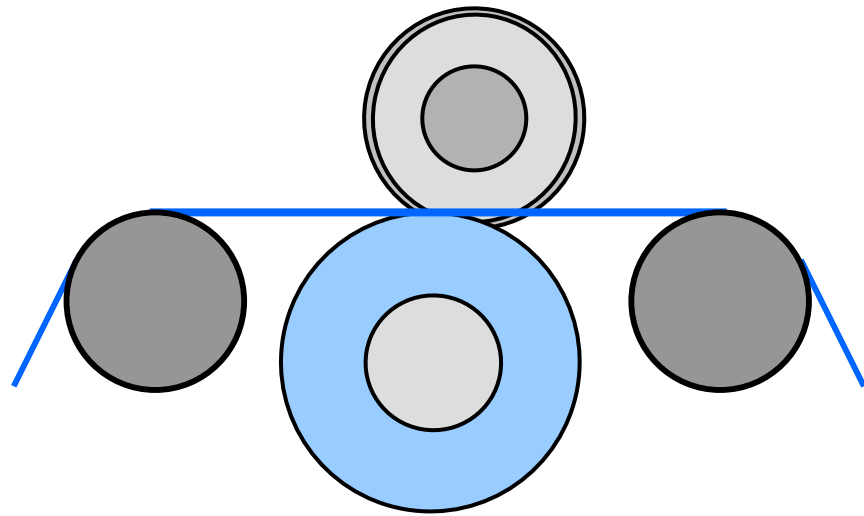
Wrap Shear



More suitable for thin,
flexible webs

Wrap curvature opposes
vertical deflection

Tangential Shear

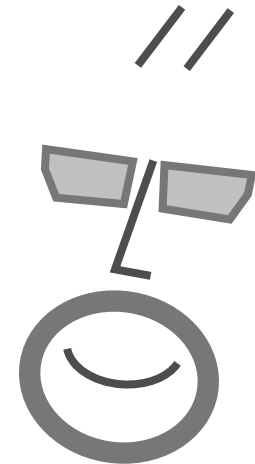


Especially suitable for thick,
more rigid webs

Flat web geometry doesn't
inhibit vertical deflection

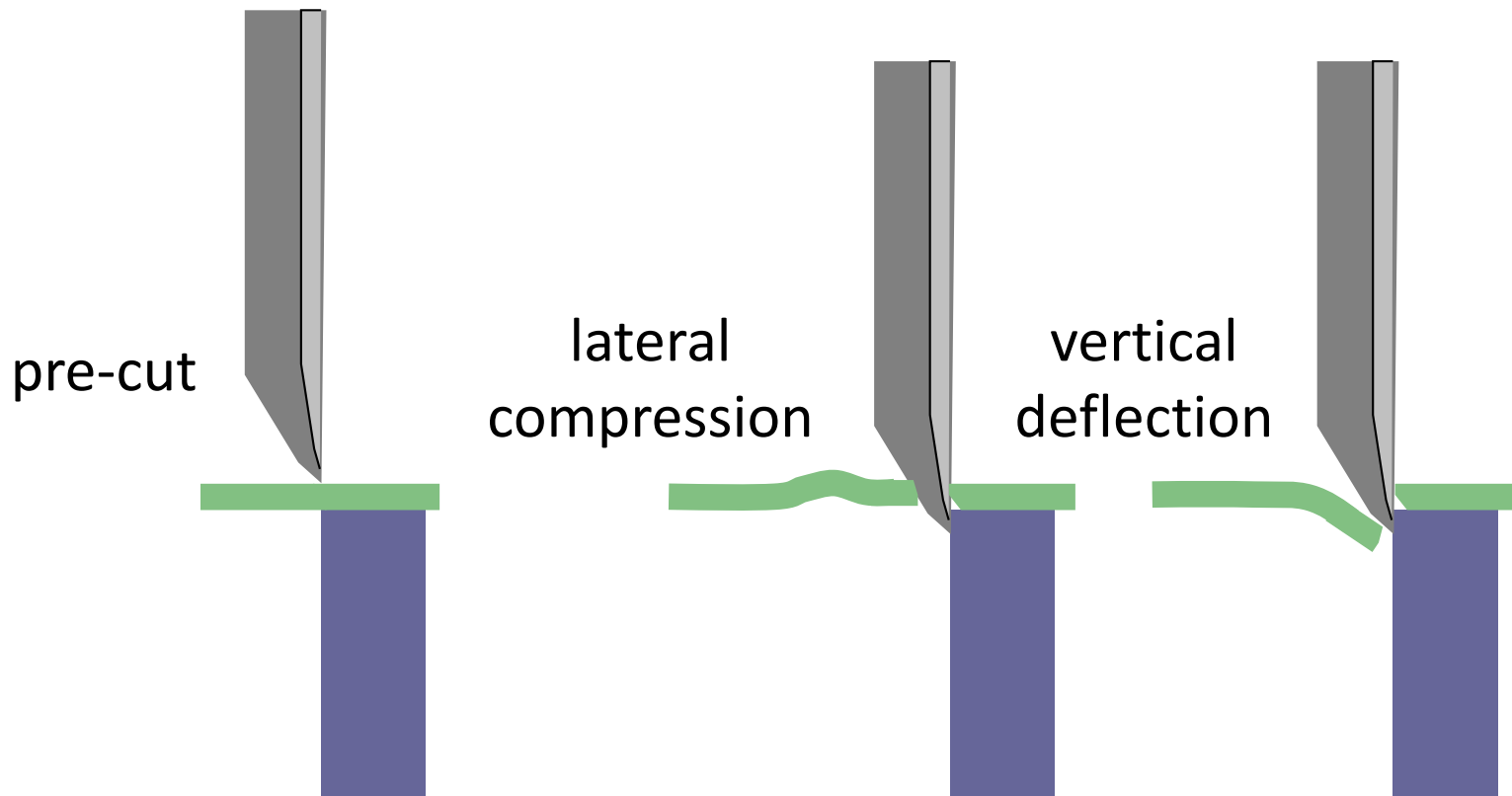
Slitting Qs

How does top shear knife setup affect shear slitting?



Passing the Top Knife

There are two extremes of how the top knife web will pass by the knife, via lateral compression or vertical deflection.



Top Shear Blade Geometry

Recommendations by Rheinhold Schable, Tidland Corp.

Schable considers
seven factors:

Caliper

Density

Elongation

Stiffness

Tensile

Abrasiveness

Compressibility

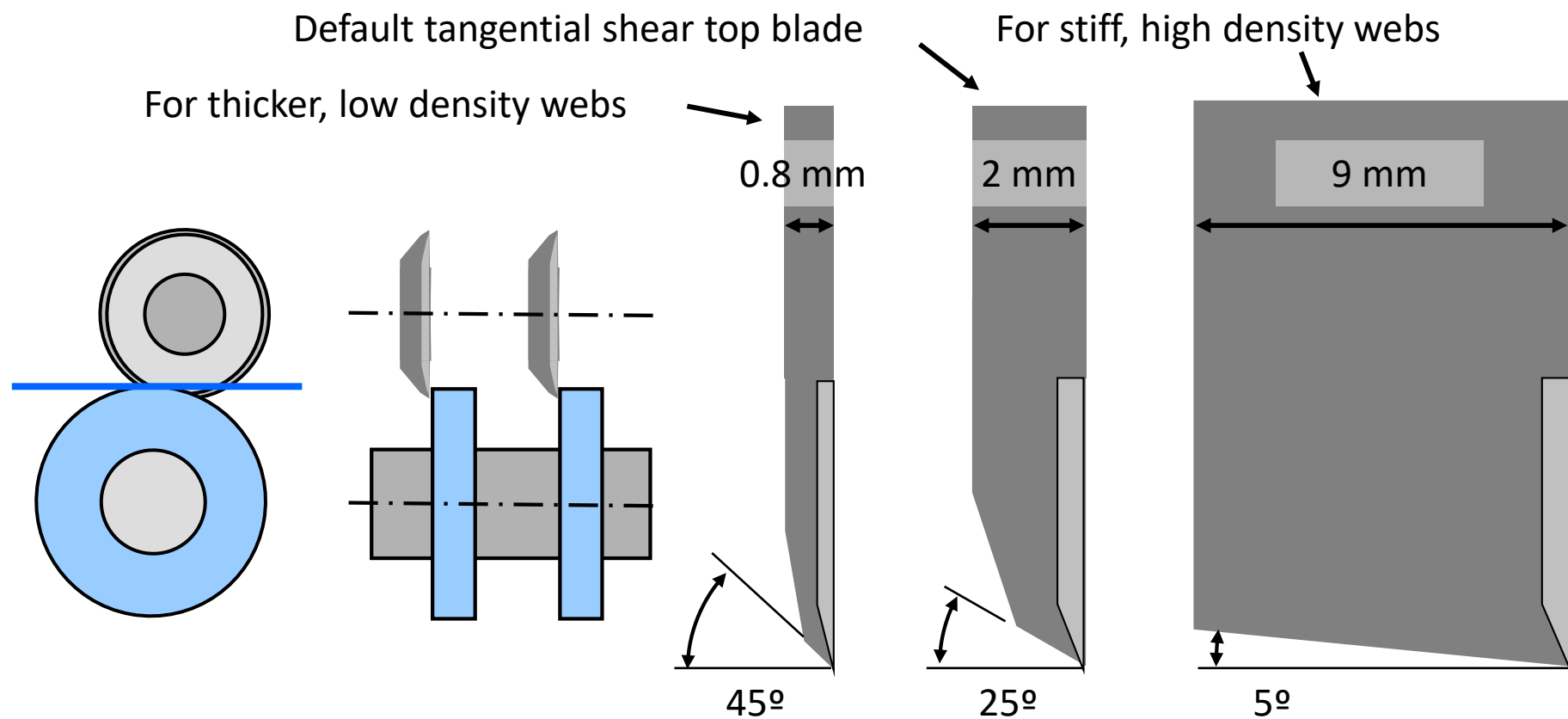
Recommended top blade setup and profile is a function of each of these factors as they vary from low to high extremes,

The key top blade parameters are:
Grind Angle,
Rim Width, and
Cant Angle

Tangential Shear Top Blade Profiles

Some typical top blade profiles for tangential shear.

Width-Angle = 0.8 mm - 45°, 2 mm - 25°, 9 mm - 5°



Top Shear Blade Geometry

General trend of recommendations:

Primary Angle = 0-10°

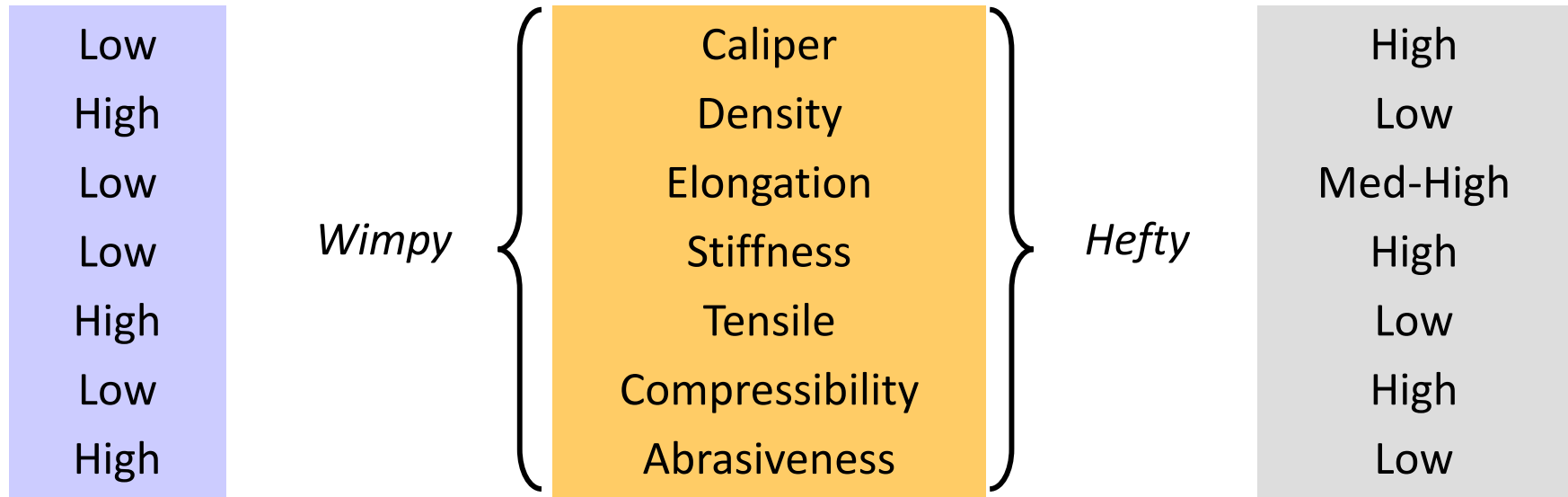
Rim Width = Wide

Cant Angle = 0-0.5°

Primary Angle = 20-45° or 45-60°

Rim Width = Narrow

Cant Angle = 0.5-1.0°



Top Blade Geometry

Back Side Face

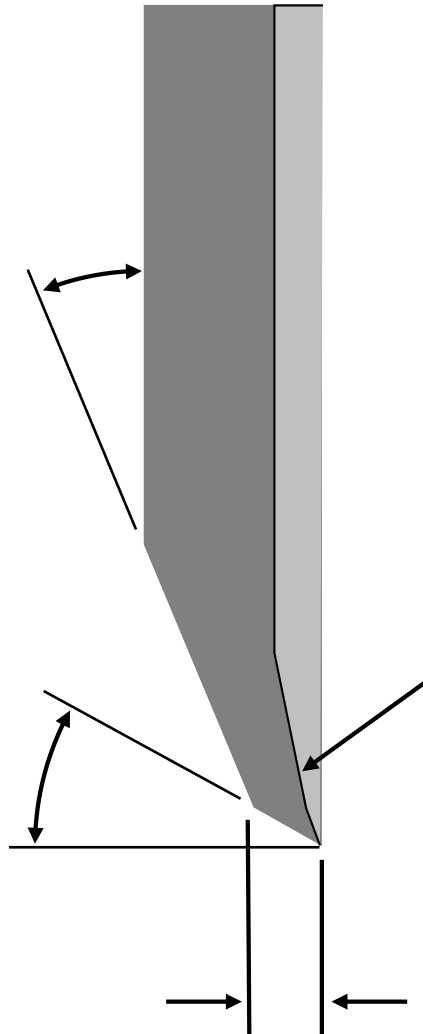
Cutting Face

Secondary
Grind Angle

Primary Grind
Angle

Undercut

Rim Width

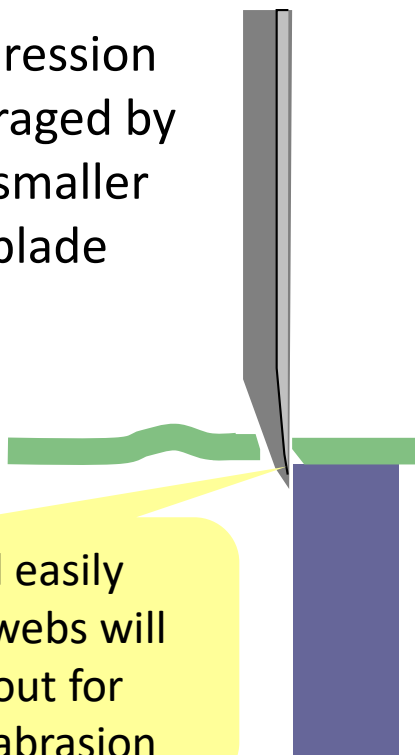


Passing the Top Knife

There are two extremes of how the top knife web will pass by the knife: lateral compression or vertical bending.

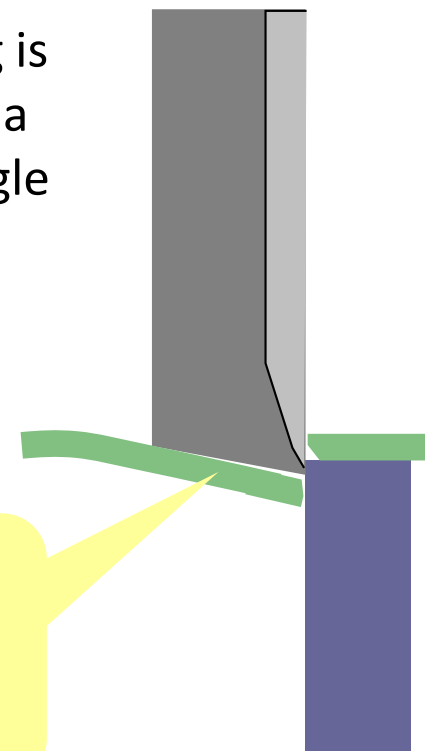
Lateral compression will be encouraged by a narrower, smaller angle top blade

Soft webs will easily compress, stiff webs will buckle. Look out for increased side abrasion



Vertical bending is encouraged by a wider, larger angle top blade.

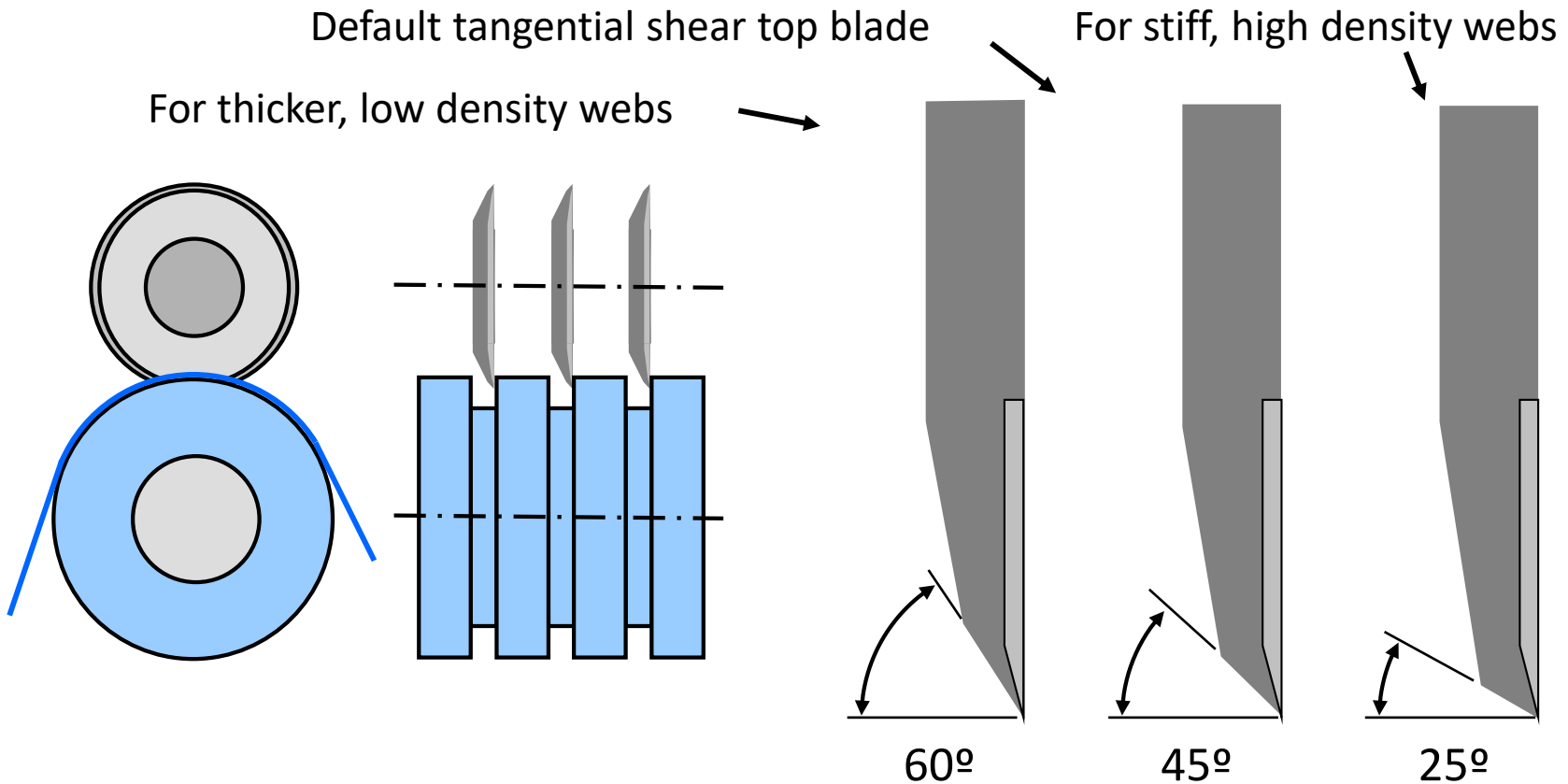
Stiffer webs will bend and recover. Look out for top face abrasion.



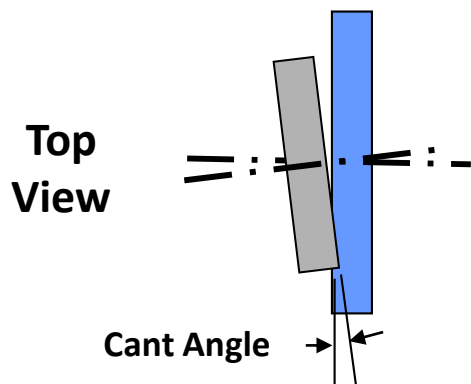
Wrap Shear Top Blade Profiles

Typical top blade profiles for wrap shear.

Width = 0.7 mm, Primary angle of 25, 45, or 60 degrees.

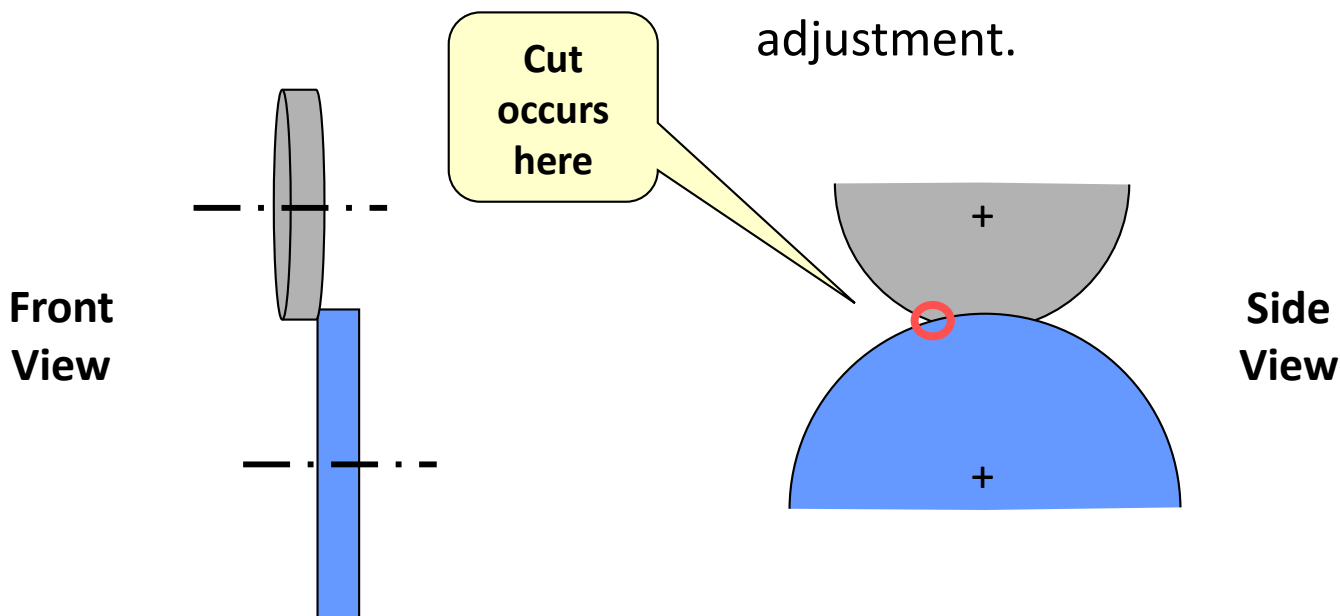


Cant Angle



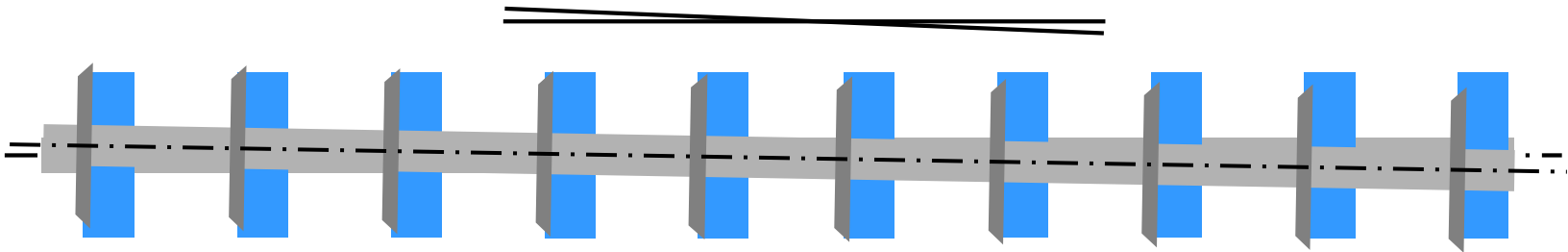
The cant angle is designed to force the shear knife contact to the overlap entrance point.

Most slitters where the top knives are all on a common shaft don't have this adjustment.

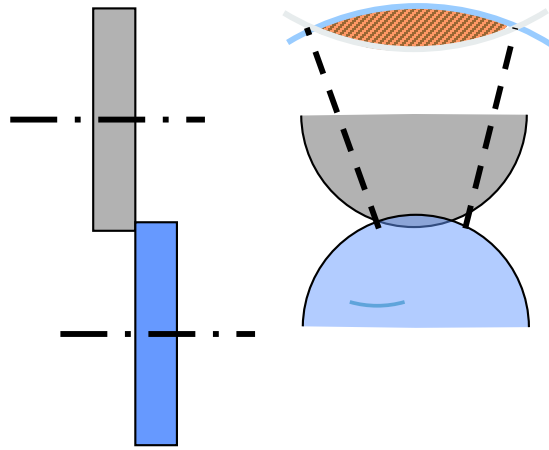


Cant Angle on Common Shaft Knives

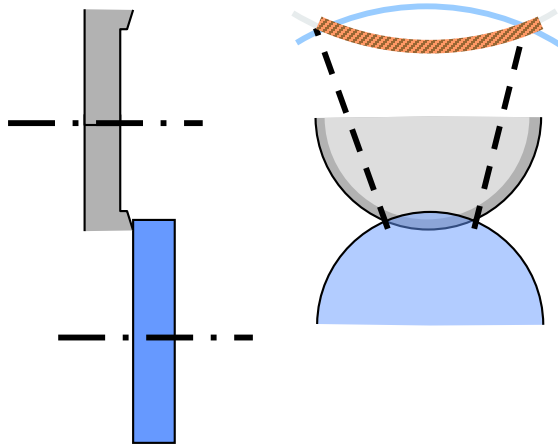
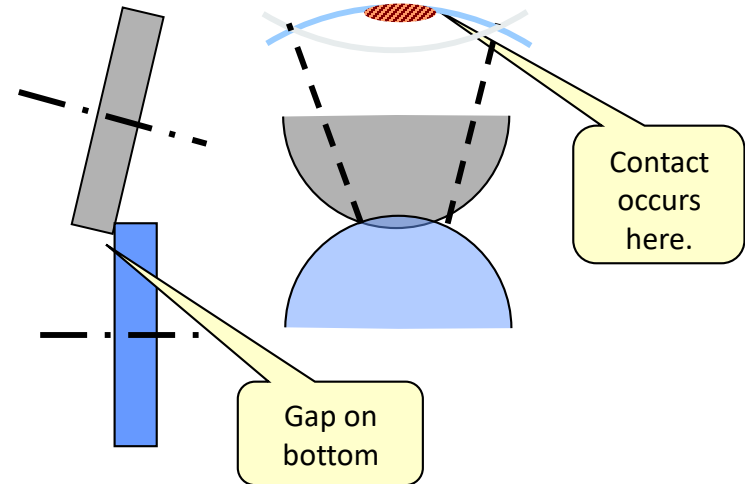
It's difficult to create a cant angle when the top knives share a common shaft.



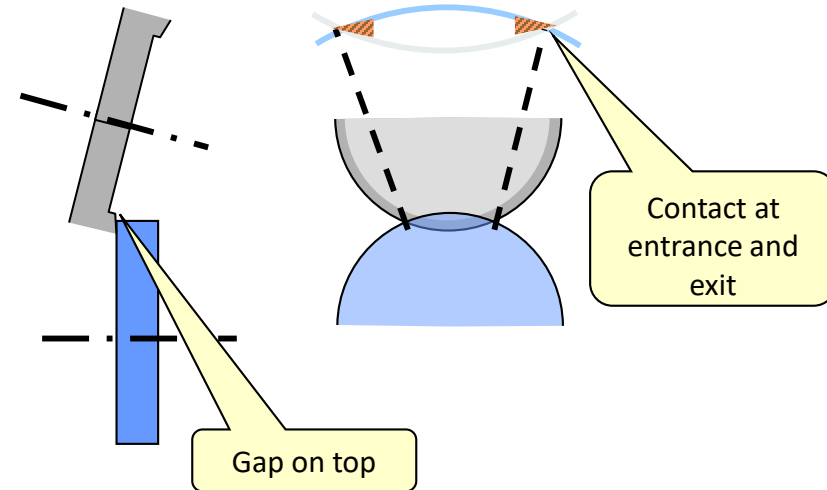
Top Knife Geometry and Tilting



With no concave angle, engagement tilting shifts contact to the high center of the overlap zone.

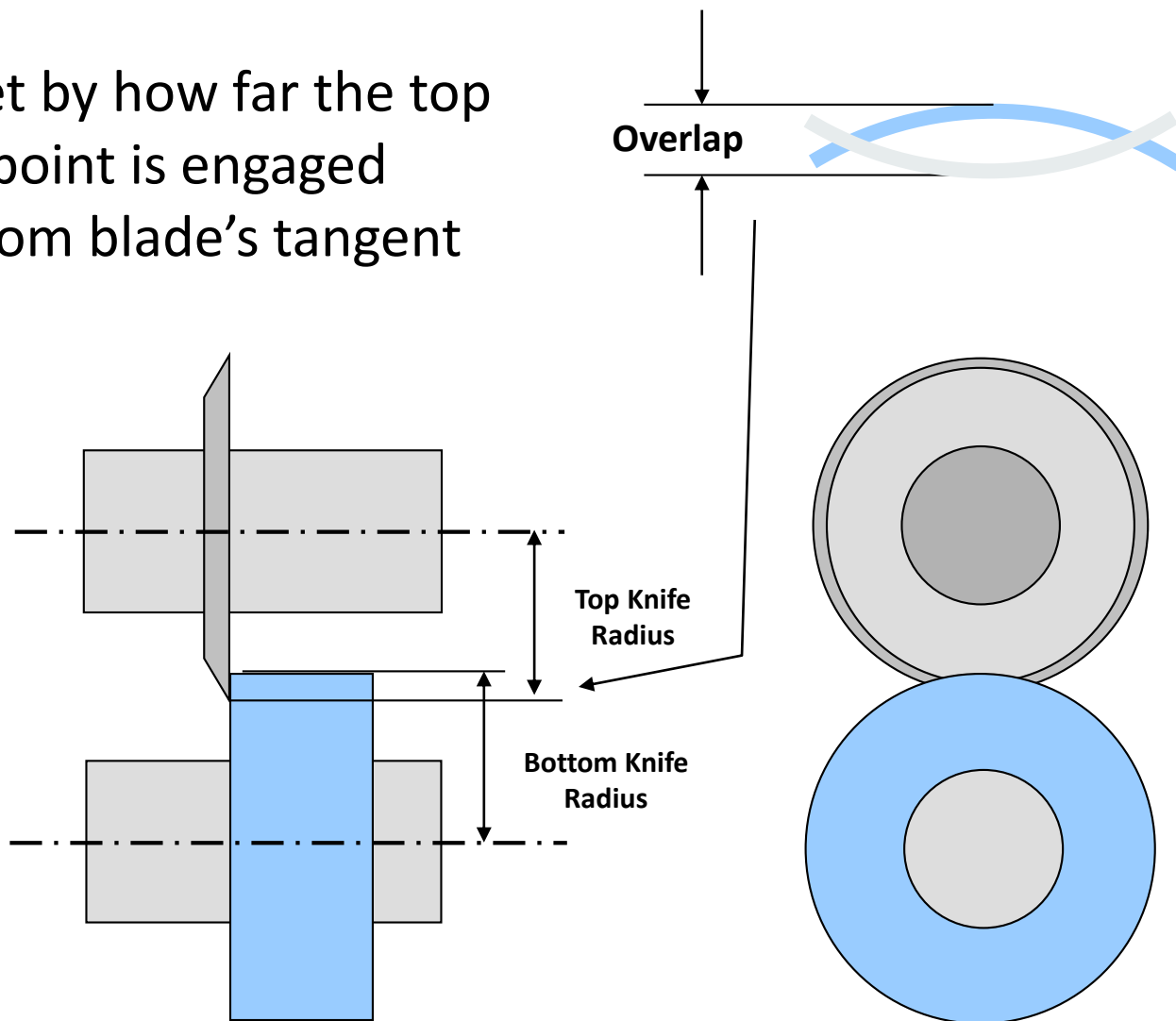


With a concave geometry the engagement tilting shifts contact to overlap entrance and exit points (as long as tilt angle is less than the concave angle).



Knife Overlap

The overlap is set by how far the top blade's tangent point is engaged beyond the bottom blade's tangent point.



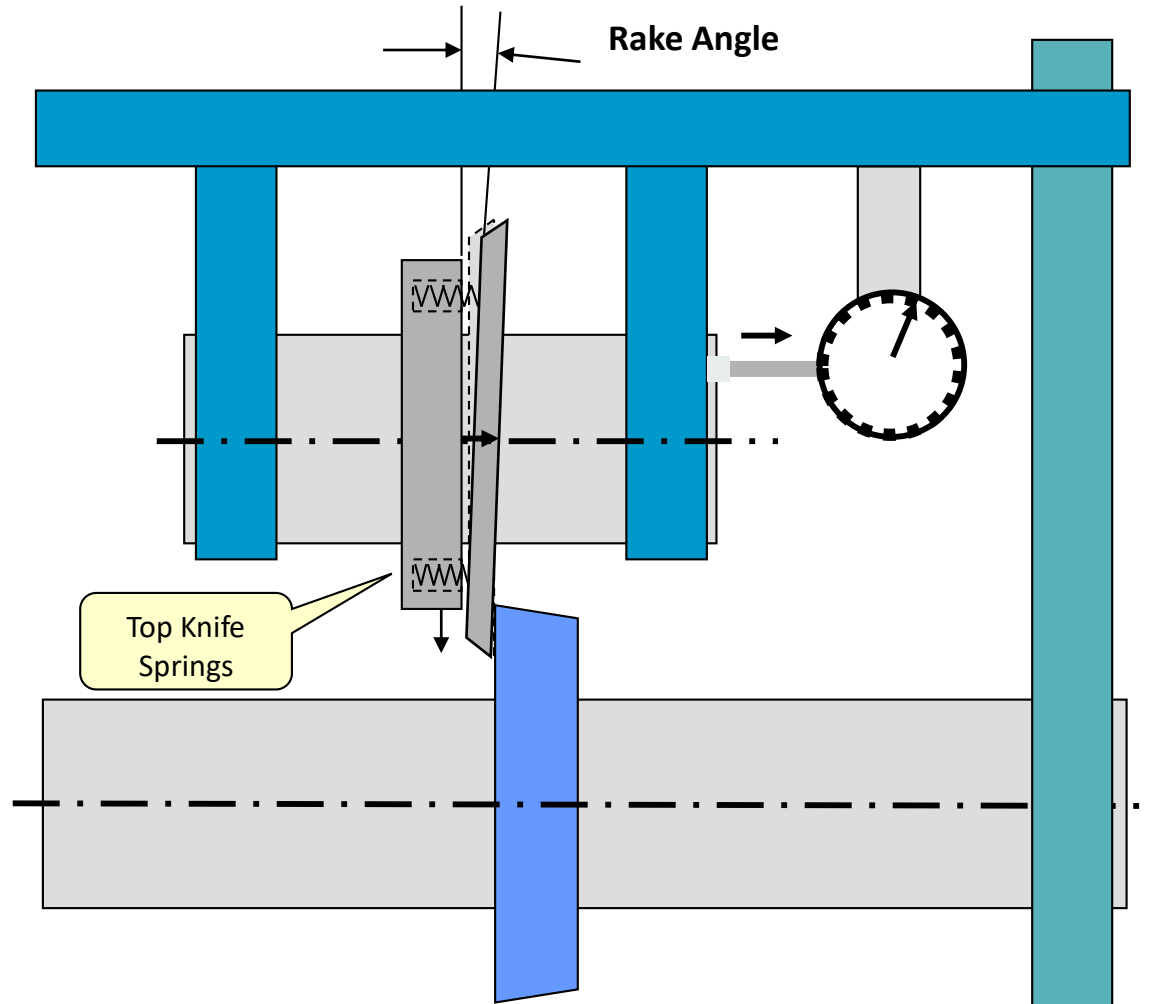
Tilt in Spring-Loaded Top Knives

The lateral load is set by adjusting the lateral shift of the top knives relative to the bottom knives.

This motion can be measured with a dial indicator.

This motion compresses the top knife springs.

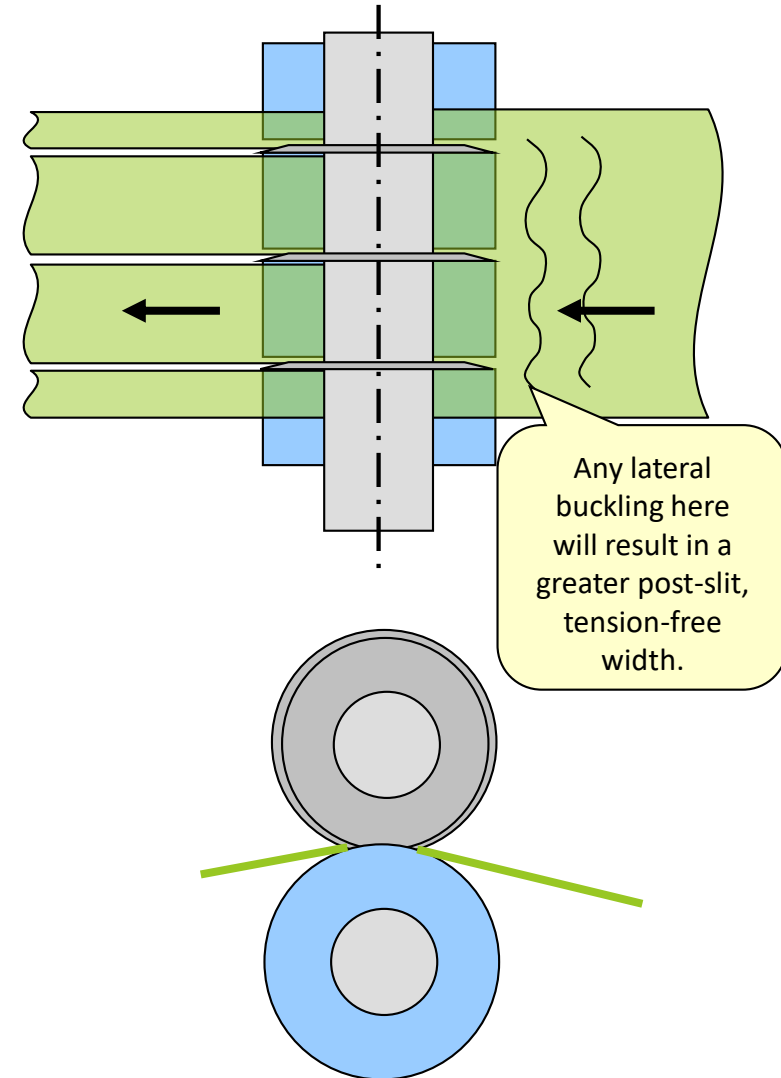
More engagement motion create more lateral force and more rake angle.



Slit Width, Knife Spacing, & Web Tilt

Knives are setup and adjusted to achieve the target web slit width. Actual slit width will be greater than knife spacing for two reasons:

1. The Poisson's effect – The web is slit under tension and recovers width when tension is removed. (This effect is extremely small with the high modulus, thick, low tension products.)
2. Buckling - If the web has any lateral buckling at the knives the post-slit, unbuckled width will be greater.

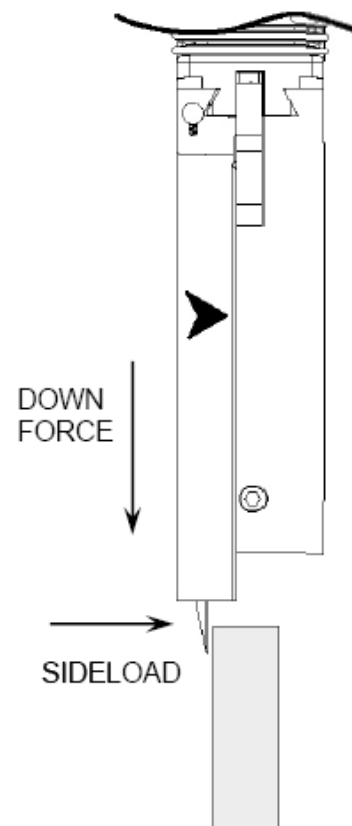
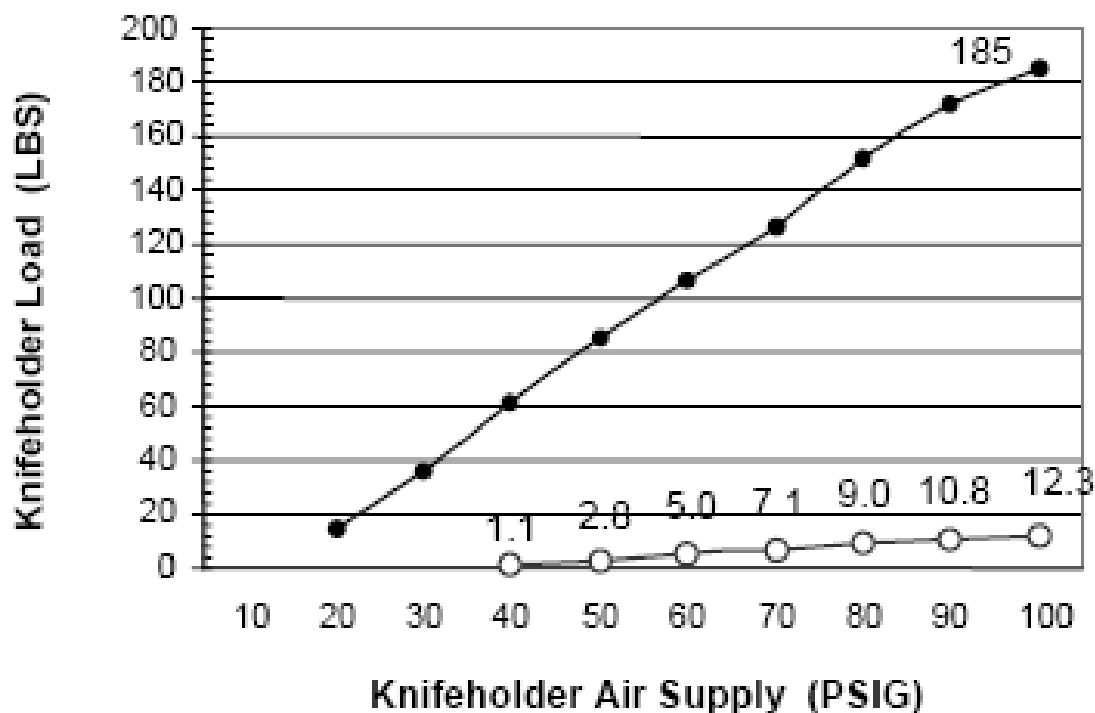


Other Shear Slitting Considerations

- Relative speed of web, bottom knife, and top knife
- Interaction of these factors with overlap and overspeed
- Setting knife overlap
- Knife wear causes

Example: Knifeholder Loads

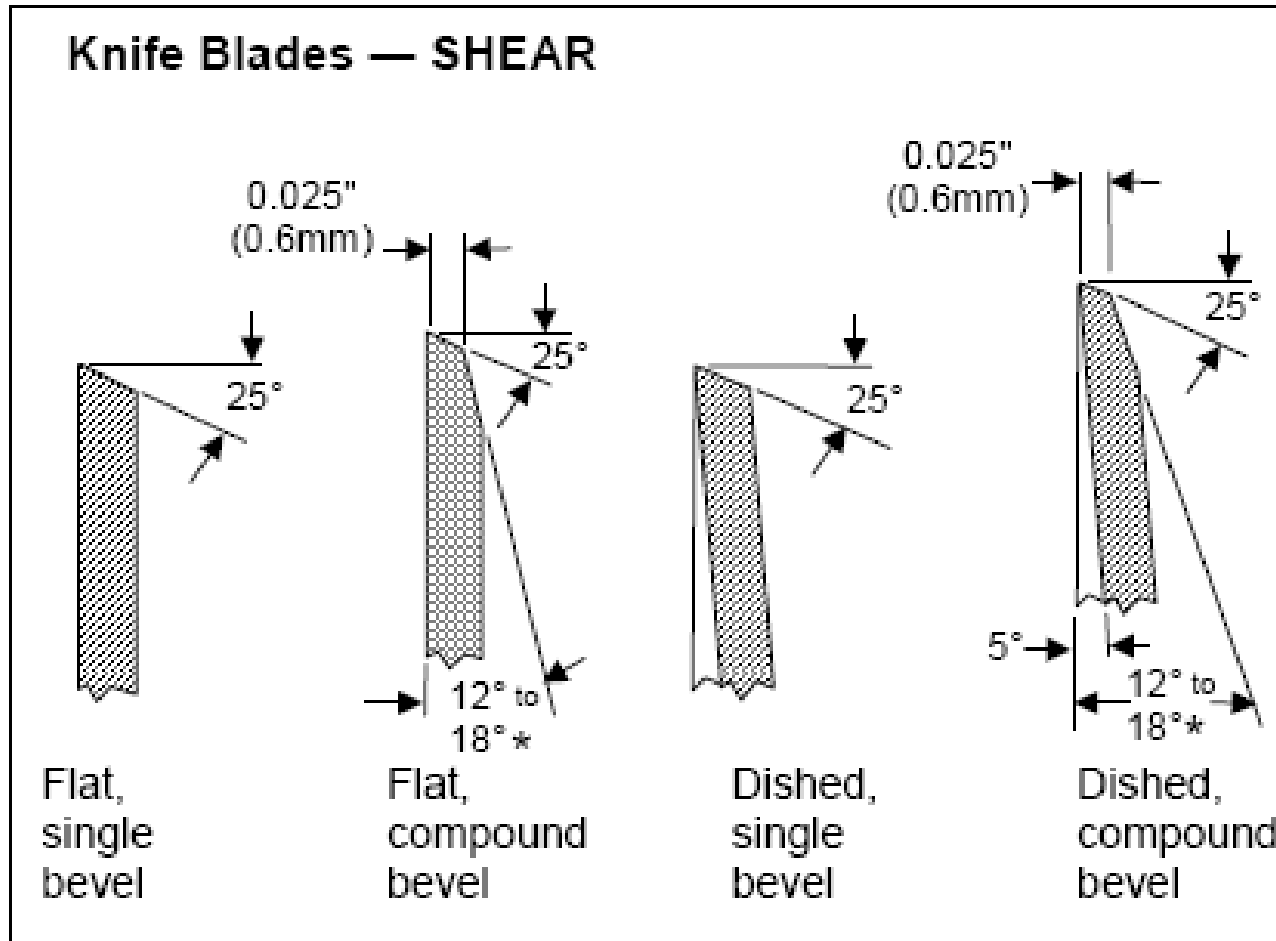
CLASS II *



Tidland CLASS I knifeholders loads are roughly $\frac{1}{2}$ of CLASS II

Image courtesy of Tidland Corp.

More on Top Shear Knife Profiles

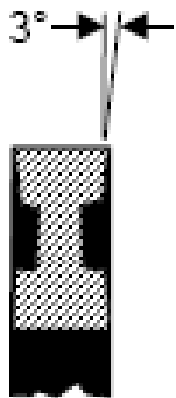


* 12° for blades $\phi 150\text{mm}$ and smaller; 18° for blades $\phi 180\text{mm}$ and larger

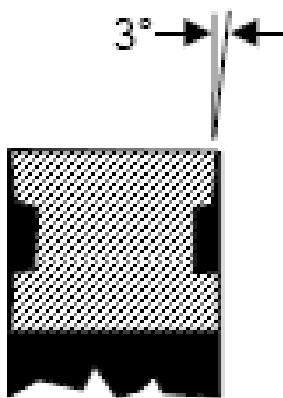
Image courtesy of Tidland Corp.

More on Bottom Shear Knife Profiles

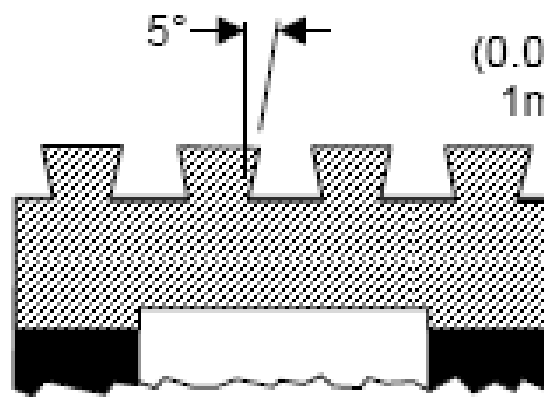
Anvil Rings — SHEAR



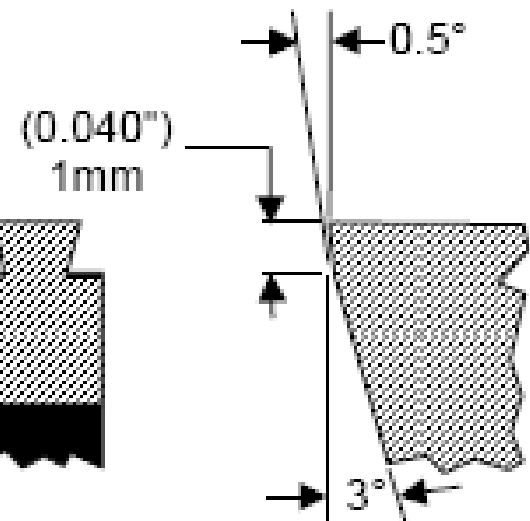
Dual edge,
all sizes



Dual edge,
all sizes



Multi-ring, custom made
to customer specifications



Carbide edge

Image courtesy of Tidland Corp.

Knife Mounting Options

Top knives on a common axis and support shaft.



Top knives with individual holders on a dovetail shaft.



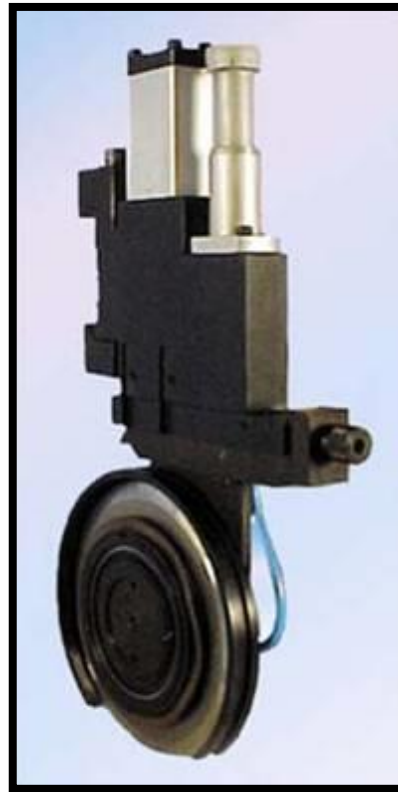
Images courtesy of Tidland Corp.

Shear Slitting on Dovetail Mount

**Top and bottom knives
on individual dovetail
mounted holders.**



**Top knives with individual holders
on a dovetail shaft.**



Images courtesy of Dienes Corp.

Auto-Positioning Shear Knives



Image courtesy of Tidland Corp.

Individual Top Knife Blade Holders

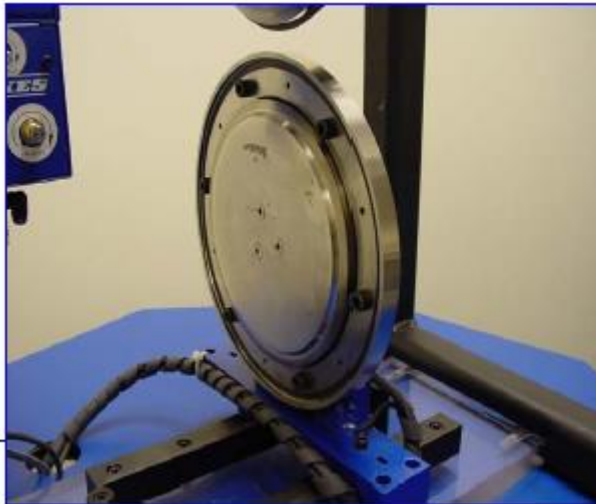


Images courtesy of Tidland Corp.

Individual Driven Bottom Shear Knife

Bottom Shear Knife Holder with Servo Motor

Images courtesy of Dienes Corp.



Razor Slitter Recommendations

For thin web, less than 1-mil (25 microns), the first choice is razor-in-groove slitting. Razor slitting tops shear slitting by its ease of setup and reduced cost of hardware and blades/sharpening.

A cleaner razor slitting system should include the following details:

- ▶ Hard coated blades to prevent loss of sharpness.
- ▶ Minimize blade width, but ensure low flexure with solid blade holders.
- ▶ Ensure a smooth side surface on razor blades.

Razor Slitter Recommendations (continued)

- ▶ Moderate to high tension to encourage Poisson's contraction away from the blade side.
- ▶ Good tension control of post-slitting webs/strands, especially trimmed edge cuts.
- ▶ Grooved roller geometry should complement razor width and engagement. Grooved roller needs minimum runout to keep a steady focal point of fracture.
- ▶ Engage either the tip of the blade or use tangentially with engagement depth less than razor bevel length.

Shear Slitting Recommendations

Shear slitting is the cleanest option for films over 2-mils thick and can be the best option for all webs when optimized properly; however, shear slitting will require more in-house expertise in knife setup and sharpening. Precision shear slitting is the most expensive option, but also the most versatile in cutting a variety of coated and laminate products. Cleaner shear slitting should include the following:

- ▶ A double-bevel, hollow ground design. Specific smoothness of all web contacting surfaces.
- ▶ Different knife geometries are needed for different web thicknesses and hardnesses.

Shear Slitting Recommendations (continued)

- ▶ Good structure will reduce deflection and allow for minimum overlap, shorter contact lengths.
- ▶ Knife sharpening needs to be supervised to ensure good profiling and avoid damage in handling.
- ▶ Wrap shear reduces contact length over kiss shear, but may not create enough benefit for the added setup time. Wrap shear will be less sensitive to web flutter and cut point shifting with overlap variations.
- ▶ Drive the top knives to minimize web-knife speed mismatch.